

Limit shapes of centrally excited random walks

Ahmed Bou-Rabee and Yuval Peres

ABSTRACT. A centrally excited random walk on $\mathbb{Z}^d$ moves like simple random walk, except that its first step from each site has a drift of fixed size toward the origin. Kozma (2007) conjectured that after n steps the visited set approximates a ball of radius of order $n^{1/(d+1)}$. We prove this in every dimension $d \geq 2$ and show that the rescaled numbers of visits converge uniformly to a cone, which is the potential generated by the drift on a ball. The same argument shows that a drift opposite to a subgradient of a norm produces the ball of that norm. In the plane, the outer and inner radii of the visited set differ by at most $n^{1/6}$ times a power of $\log n$, and the exponent $1/6$ cannot be lowered.

1. INTRODUCTION

Let $d \geq 2$ and $0 < \varepsilon < 1/d$. A **centrally excited random walk** $X_0 = 0, X_1, X_2, \dots$ on $\mathbb{Z}^d$ moves like simple random walk, except that on its first departure from each site $x \neq 0$ it moves to $x \pm e_i$ with probability $1/(2d) \mp \varepsilon x_i/(2|x|)$ for $1 \leq i \leq d$, where $e_1, \dots, e_d$ is the standard basis of $\mathbb{R}^d$. Given the path up to that departure, this step has mean $-\varepsilon x/|x|$, a drift of size ε toward the origin (Figure 1).

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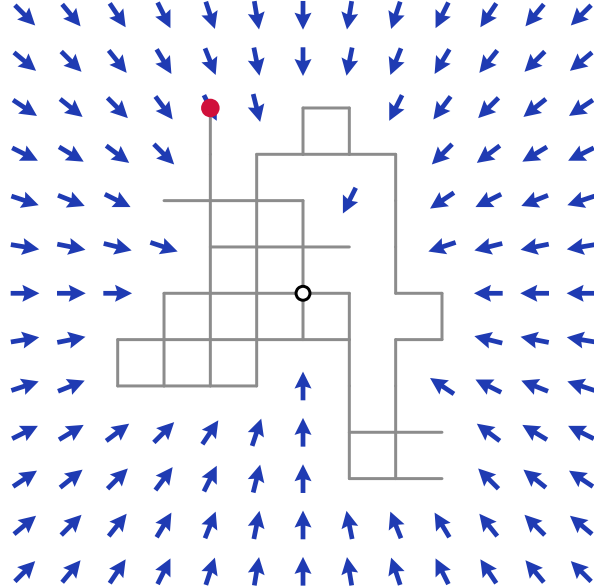


FIGURE 1. The walk after 120 steps, for $d = 2$ and $\varepsilon = 0.4$: its path (gray), the origin (open circle) and its position (red dot). At each site $x \neq 0$ that it has not yet left, an arrow shows the mean $-\varepsilon x/|x|$ of its first step from x .

At an Oberwolfach problem session, [Kozma \[2007\]](#) conjectured that the visited set approximates a ball of radius of order $n^{1/(d+1)}$. We prove this for every $d \geq 2$, identify the radius and the uniform limit of the local times, and bound their fluctuations. In the plane the radii fluctuate on the scale $n^{1/6}$ up to logarithmic factors; in higher dimensions they are bounded by powers of $\log n$. More generally, a first-departure drift opposite to a subgradient of a norm produces the ball of that norm.

1.1. The limit shape. For an integer $n \geq 0$ let

$$\ell_n(x) := \sum_{j=0}^{n-1} \mathbf{1}_{\{X_j=x\}} \quad \text{and} \quad A_n := \{x : \ell_n(x) > 0\}.$$

The **local time** $\ell_n(x)$ is the number of departures from x before time n , and A_n is the **range**; the set of sites visited by time n is A_{n+1} . Write $B(y, \rho)$ for the open Euclidean ball of radius ρ about y and ω_d for the volume of $B(0, 1)$, and let

$$(1.1) \quad r_n := \left(\frac{(d+1)n}{2d\varepsilon\omega_d} \right)^{\frac{1}{d+1}},$$

so that the cone $2d\varepsilon(r_n - |v|)_+$ has integral n . The theorem below says that the range approximates $B(0, r_n)$ and the local times approximate this cone. Its slope $2d\varepsilon$ comes from the potential of a ball, computed in (1.10); its integral determines r_n .

Theorem 1.1 (Limit shape). *Let $d \geq 2$ and $0 < \varepsilon < \frac{1}{d}$. Then, almost surely, the centrally excited random walk has the following properties.*

(i) Shape: for every $0 < \eta < 1$ and all sufficiently large n ,

$$\{x \in \mathbb{Z}^d : |x| < (1 - \eta)r_n\} \subset A_n \subset \{x \in \mathbb{Z}^d : |x| < (1 + \eta)r_n\}.$$

(ii) Local times: as $n \rightarrow \infty$,

$$\frac{1}{r_n} \max_{x \in \mathbb{Z}^d} |\ell_n(x) - 2d\varepsilon(r_n - |x|)_+| \rightarrow 0.$$

(iii) Recurrence: the walk visits every site of $\mathbb{Z}^d$ infinitely often.

Part (ii) implies recurrence and supplies the bracket limits used in Section 8. Part (i) gives $|A_n|/r_n^d \rightarrow \omega_d$, and Theorem 1.2 gives rates.

1.2. Fluctuations. Replace each site $x \in A_n$ by the unit cell $C_x := x + [-\frac{1}{2}, \frac{1}{2})^d$, and let $D_n := \bigcup_{x \in A_n} C_x$. The **inner radius** and the **outer radius** are

$$(1.2) \quad R_{\text{in}}(n) := \inf_{y \notin D_n} |y| \quad \text{and} \quad R_{\text{out}}(n) := \max_{0 \leq j \leq n} |X_j|.$$

With high probability the radii differ from r_n by at most a power of $\log n$ in $d \geq 3$, and by at most $n^{1/6}$ times a power of $\log n$ in the plane.

Theorem 1.2 (Fluctuation bounds). *Let $d \geq 2$ and $0 < \varepsilon < 1/d$, and consider the centrally excited random walk. For every $p > 0$ there exists $C(d, \varepsilon, p) < \infty$ such that, for every integer $n \geq 2$, the following bounds hold simultaneously with probability at least $1 - Cn^{-p}$.*

(i) Radii:

$$\begin{aligned} |R_{\text{in}}(n) - r_n| &\leq C\sqrt{r_n \log n} \quad \text{and} \quad R_{\text{out}}(n) - r_n \leq C\sqrt{r_n}(\log n)^{5/2} \quad \text{if } d = 2, \\ |R_{\text{in}}(n) - r_n| &\leq C \log n \quad \text{and} \quad R_{\text{out}}(n) - r_n \leq C(\log n)^{d+1} \quad \text{if } d \geq 3. \end{aligned}$$

(ii) Volume:

$$|r_n^{-1}D_n \triangle B(0,1)| \leq C \begin{cases} \sqrt{\frac{\log n}{r_n}}, & d = 2, \\ \frac{\log n}{r_n}, & d \geq 3 \end{cases}.$$

(iii) Local times:

$$\max_{x \in \mathbb{Z}^d} |\ell_n(x) - 2d\varepsilon(r_n - |x|)_+| \leq C \begin{cases} \sqrt{r_n}(\log n)^{3/2}, & d = 2, \\ \sqrt{r_n \log n}, & d \geq 3 \end{cases}.$$

Moreover, almost surely these bounds hold for all sufficiently large n .

Since $|D_n| = |A_n|$, the volume bound also controls $||A_n|/r_n^d - \omega_d|$. The next theorem shows that the planar exponent $1/6$ cannot be lowered and that the local-time bound has the correct order in $d \geq 3$.

Theorem 1.3 (Sharpness). *Let $d \geq 2$ and $0 < \varepsilon < 1/d$, and consider the centrally excited random walk.*

(i) Radii in the plane: *let $d = 2$. For every $p > 0$ there exists $c(\varepsilon, p) > 0$ such that, for all sufficiently large n ,*

$$(1.3) \quad \mathbb{P}(r_n - R_{\text{in}}(n) \geq c\sqrt{r_n \log n}) \geq n^{-p} \quad \text{and} \quad \mathbb{P}(R_{\text{out}}(n) - r_n \geq c\sqrt{r_n \log n}) \geq n^{-p}.$$

(ii) Iterated logarithm for the radii in the plane: *let $d = 2$. Almost surely,*

$$(1.4) \quad \limsup_{n \rightarrow \infty} \frac{r_n - R_{\text{in}}(n)}{\sqrt{r_n \log \log n}} \geq \frac{1}{\sqrt{10\pi\varepsilon}} \quad \text{and} \quad \limsup_{n \rightarrow \infty} \frac{R_{\text{out}}(n) - r_n}{\sqrt{r_n \log \log n}} \geq \frac{1}{\sqrt{10\pi\varepsilon}}.$$

(iii) Local times near the origin: *there exists $c(d, \varepsilon) > 0$ such that, for all sufficiently large n , with probability at least $1 - n^{-c}$,*

$$(1.5) \quad \max_{\substack{x \in \mathbb{Z}^d \\ |x| \leq r_n^{1/2}}} |\ell_n(x) - 2d\varepsilon(r_n - |x|)| \geq c \begin{cases} \sqrt{r_n \log n}, & d = 2, \\ \sqrt{r_n \log n}, & d \geq 3 \end{cases}.$$

(iv) Difference of the radii in the plane: *let $d = 2$.*

(a) *For every $p > 0$ there exists $c(\varepsilon, p) > 0$ such that, for all sufficiently large n ,*

$$(1.6) \quad \mathbb{P}(R_{\text{out}}(n) - R_{\text{in}}(n) \geq c\sqrt{r_n \log n}) \geq n^{-p}.$$

(b) *Almost surely,*

$$(1.7) \quad \limsup_{n \rightarrow \infty} \frac{R_{\text{out}}(n) - R_{\text{in}}(n)}{\sqrt{r_n \log \log n}} \geq \sqrt{\frac{\pi}{3\varepsilon}}.$$

(c) *For every $a \geq 0$,*

$$(1.8) \quad \limsup_{n \rightarrow \infty} \mathbb{P}(R_{\text{out}}(n) - R_{\text{in}}(n) \leq a\sqrt{r_n}) \leq 1 - \exp\left(-\frac{3\varepsilon a^2}{\pi}\right).$$

Parts (i) and (iv)(a) rule out bounds of order $o(\sqrt{r_n \log n})$ on $|R_{\text{in}}(n) - r_n|$ and on $R_{\text{out}}(n) - R_{\text{in}}(n)$ with failure probability $O(n^{-p})$ for a fixed $p > 0$. Part (iv)(c), together with Theorem 1.2, gives the scale of the width in probability: for every $\eta > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(n^{1/6-\eta} \leq R_{\text{out}}(n) - R_{\text{in}}(n) \leq n^{1/6+\eta}) = 1.$$

The upper bound in the last display also holds almost surely for all large n ; the corresponding almost-sure lower bound is open. Parts (i) and (ii) do not control the width, since both radii could move together. Part (iv) instead uses the sum of the vectors $x/|x|$ over the range: the terms inside

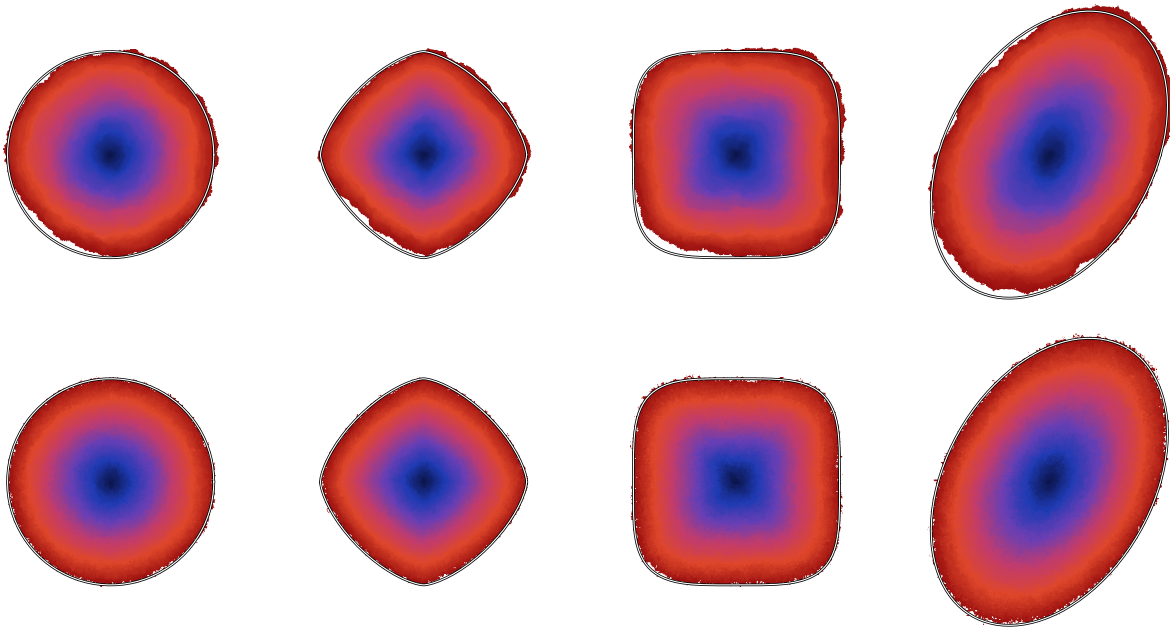


FIGURE 2. Ranges $r_n^{-1}A_n$ for the Euclidean, $\ell^{3/2}$, ℓ^4 and elliptical norms (left to right), at $n = 10^9$. Top: $d = 2$, $\varepsilon = 0.4$; bottom: $x_3 = 0$ sections for $d = 3$, $\varepsilon = 0.25$. Curves mark unit balls; blue-to-red colors show $(t_x/n)^{1/(d+1)}$, with t_x the first-departure time.

the inner disk cancel, leaving only sites between the two radii. Near the origin, Proposition 9.2 matches the local-time lower bound in part (iii), and almost surely this lower bound holds at times $n = 2^k$ for all large k . Proposition 9.4 also gives $\max\{r_n - R_{\text{in}}(n), R_{\text{out}}(n) - r_n\} > c \log n$ eventually, and $R_{\text{out}}(n) - r_n > c \log n$ in $d \geq 3$. The optimal logarithmic powers in the outer-radius bound remain open.

At a fixed site, the local-time deviation satisfies a central limit theorem and a law of the iterated logarithm, with normalization $\sqrt{r_n \log r_n}$ in the plane and $\sqrt{r_n}$ in $d \geq 3$ (Theorem 8.3).

1.3. Limit shapes for other norms. Let Ψ be a norm with unit ball B_Ψ . Replace $x/|x|$ in the first-departure probabilities by a fixed subgradient $\xi(x)$ of Ψ at x , and assume $\varepsilon \max_i \Psi(e_i) < 1/d$. The step then has mean $-\varepsilon \xi(x)$. Almost surely the range approximates $\{\Psi < r_n\}$ and the local times approximate $2d\varepsilon(r_n - \Psi)_+$, with $|B_\Psi|$ replacing ω_d in (1.1) (Theorem 2.1). Here a subgradient satisfies $\Psi(y) \geq \Psi(x) + \xi(x) \cdot (y - x)$ for every y ; at a differentiability point it is the gradient.

The theorem includes norms with corners, such as $\|\cdot\|_1$ and $\|\cdot\|_\infty$, with rates uniform in the choice of subgradients. It also extends to the Minkowski functional of every compact convex set containing the origin in its interior (end of Section 5). Figures 2 and 3 show the range and the local times. In all eight range simulations, $(|A_n|/|B_\Psi|)^{1/d}$ differs from r_n by less than one percent. The elliptical balls have semi-axes $3/2, 1$ or $3/2, 1, 1$, with major axis parallel to $(1, \sqrt{3})$ or $(1, \sqrt{3}, 0)$; t_x is the first-departure time from x .

1.4. Strategy and key ideas. The proof rests on an explicit potential. For a bounded measurable set $D \subset \mathbb{R}^d$ and $y \in \mathbb{R}^d$, let

$$(1.9) \quad U_D(y) := \frac{2\varepsilon}{\omega_d} \int_D \frac{v}{|v|} \cdot \frac{v - y}{|v - y|^d} dv$$

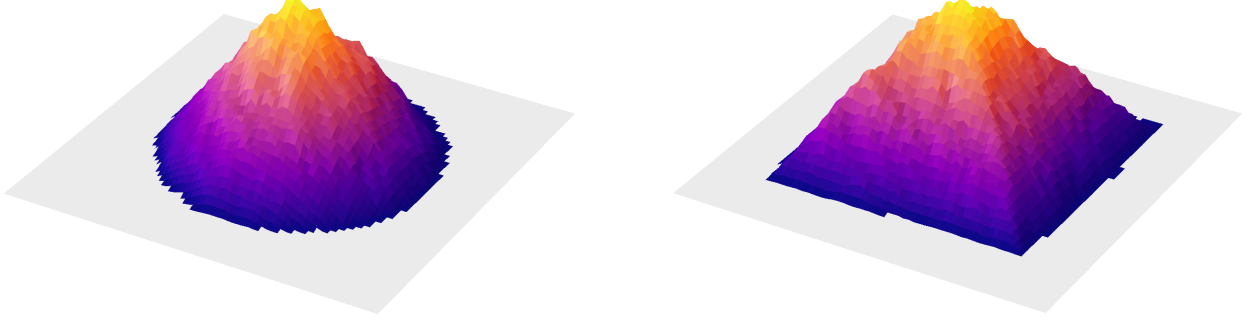


FIGURE 3. Local times $\ell_n(x)/(2d\varepsilon r_n)$ over x/r_n for the Euclidean norm (left) and ℓ^∞ (right): $d = 2$, $n = 10^{10}$, $\varepsilon = 0.4$. Heights average 75×75 and 69×69 blocks, respectively; gray marks unvisited blocks.

be the potential generated by the vector field $-\varepsilon v/|v|$ on D . Integration along the rays from y shows that the potential of a ball is a cone: for every $\rho > 0$ and every $y \in \mathbb{R}^d$,

$$(1.10) \quad U_{B(0,\rho)}(y) = 2d\varepsilon(\rho - |y|)_+.$$

Replacing $v/|v|$ by $\nabla\Psi(v)$ gives the cone $2d\varepsilon(\rho - \Psi(y))_+$ for a ball of the norm (Lemma 2.2).

Dynkin's formula compares the local times with U_{D_n} , up to a martingale and a smaller error. The obstacle is a thin extension of the range: its potential can be smaller than the martingale error. We combine two facts to exclude it. Local times vanish outside the range, and a walk can reach a distant site only by crossing the intervening region against its first-departure drift. The decisive comparison is at a contact point of the largest ball about the origin contained in D_n : the potential is small there, and the excess outside the ball contributes nonnegatively.

1. *Local times and the potential* (Section 3). Freedman's inequality controls the Dynkin martingales uniformly over sites and time intervals. Replacing the discrete drift sum by U_{D_n} costs $O(\log n)$, including for norms with corners; convexity controls the variation of subgradients across a cell (Lemma 3.3).
2. *Crossing and coarse bounds* (Section 4). The inward drift prevents the walk from crossing a region with few departures. This bounds its radius by the largest local time outside a ball. A first application gives a crude radius bound; the contact comparison then bounds those exterior local times, and a second application gives $R_{\text{out}}(n) \asymp \max_x \ell_n(x) \asymp r_n$ and $|A_n| \asymp r_n^d$ (Proposition 4.1).
3. *The limit shape* (Section 5). At a contact point y_0 , Dynkin's formula at a neighboring site outside A_n bounds $U_{D_n}(y_0)$ (Lemma 5.2). Convexity makes the contribution of every excess point v at least $c(\Psi(v) - b)r_n^{-d}$, where b is the radius of the largest norm ball in D_n . The excess therefore has small potential, giving the conical local times; the quadratic martingale gives $b \approx r_n$. To exclude thin extensions for a norm with corners, we cross in the direction of the Euclidean projection onto the norm ball, whose normal has positive inner product with every subgradient at sites near the maximum in that direction.
4. *The inner radius* (Section 6). For the Euclidean norm, an excess point v contributes at least $c|v - y_0|^{2-d}/r_n$ at the contact point. This bounds the excess volume by $Cr_n^{d-1}U_{D_n}(y_0)$ and sharpens the inner-radius estimate (Proposition 6.1).
5. *The outer radius* (Section 7). We evaluate the potential beyond the outer radius, where the local time vanishes. The excess near the enclosing sphere contributes negatively; curvature bounds the positive contribution. These volume bounds control the local times outside the inner ball, and the crossing estimate gives the outer-radius bound (Proposition 7.1). The same estimates give the uniform local-time bound.

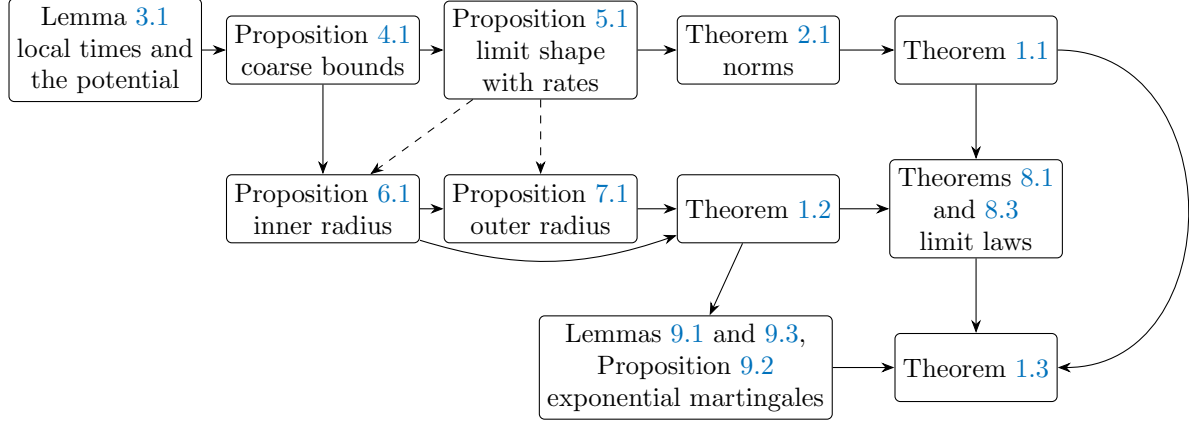


FIGURE 4. How the main results depend on each other; every result also uses Lemma 3.1. Propositions 6.1 and 7.1 concern the Euclidean norm, and the dashed arrows mean that Proposition 6.1 uses Lemma 5.2 and Proposition 7.1 uses Lemmas 3.4 and 4.3, with other estimates of Sections 4 and 5, but neither uses the conclusion of Proposition 5.1. Theorem 1.3 (i) uses Theorems 1.2 and 8.1 and Lemma 9.1, part (ii) uses Theorem 8.1, part (iii) uses Proposition 9.2 and Lemmas 9.1 and 9.3, and part (iv) uses Lemma 9.1, Proposition 4.1 and Theorem 1.1 (i).

The conical profile determines the bracket limits, so the martingale central limit theorem and the law of the iterated logarithm give the limit laws of Section 8. Exponential martingales give the larger deviations needed for the lower bounds: one martingale controls the planar radii, and many weakly correlated martingales control the largest local time. The width uses the vector martingale $X_n + \varepsilon \sum_{x \in A_n} x/|x|$. Cancellation inside the inner disk turns its order $\sqrt{n}$ fluctuations into width fluctuations of order $\sqrt{r_n}$; its two-dimensional Gaussian limit gives the small-ball bound. Figure 4 records the dependencies.

The local times play the role of the divisible-sandpile odometer, compared with an explicit potential in [Levine and Peres \[2009\]](#). Martingales with a pole at an unvisited site and separate estimates against thin extensions also appear in the internal diffusion-limited aggregation bounds of [Jerison, Levine, and Sheffield \[2012, 2013\]](#). Our logarithmic lower bounds use the inner/outer deviation dichotomy of [Asselah and Gaudillière \[2014, Proposition 1.2\]](#).

1.5. Related work. [Kozma \[2013\]](#) reported recurrence of the walk excited to the center, citing unpublished work from 2006. [Dembo, Huang, and Sidoravicius \[2014a\]](#) described the manuscript of [Kozma \[2006\]](#) for a related walk in $d \geq 3$, with drift $-\varepsilon x/\|x\|_1$ at the first visit to a site having an unvisited neighbor; [Dembo, Huang, and Sidoravicius \[2014b\]](#) cited it for recurrence in every dimension. We take the name centrally excited random walk from its title. Its direction-dependent drift magnitude differs from our constant Euclidean drift. [Benjamini, Duminil-Copin, Kozma, and Lucas \[2020\]](#) also recorded the predicted radius $t^{1/(d+1)}$.

For the excited random walk with first-visit drift in a fixed direction, [Benjamini and Wilson \[2003\]](#) proved transience in $d \geq 2$ and positive lower speed in $d \geq 4$. [Kozma \[2003, 2005\]](#) obtained positive lower speed in $d = 3$ and $d = 2$, and [Bérard and Ramírez \[2007\]](#) proved laws of large numbers and central limit theorems. [Menshikov, Popov, Ramírez, and Vachkovskaia \[2012\]](#) extended positive-speed results to uniformly elliptic walks with bounded jumps and uniformly positive first-visit drift in a fixed direction and zero drift at later visits. [Zerner \[2005, 2006\]](#) gave recurrence/transience criteria on $\mathbb{Z}$ and strips for walks with multiple cookies. [Amir, Benjamini, and Kozma \[2008\]](#) proved recurrence for a walk in the upper half of $\mathbb{Z}^3$ that steps down at new sites. See [Kosygina and Zerner \[2013\]](#) for a survey.

The radius $n^{1/(d+1)}$ and range size $n^{d/(d+1)}$ are also predicted for once-reinforced walks with strong reinforcement [Beffara, 2011; Elboim and Kozma, 2026], and supported by simulations [Foster, Grassberger, and Paczuski, 2009]; related predictions for self-attracting walks appear in Ordemann, Berkolaiko, Havlin, and Bunde [2000]; Ordemann, Tomer, Berkolaiko, Havlin, and Bunde [2001]. For once-reinforced random walk on $\mathbb{Z}^d$, $d \geq 2$, Bou-Rabee and Peres [2026b] proved the lower bound $cn^{d/(d+1)}$ on the expected range for every reinforcement parameter. The matching upper bound and limit shape remain conjectural. See Pemantle [2007] for reinforced processes.

Dembo, Groisman, Huang, and Sidoravicius [2021] proved an averaging principle and shape theorem for growth models motivated by origin-excited and once-reinforced walks. Their simulations with inward first steps along a coordinate of largest absolute value or along all coordinates produced squares and diamonds. Our norm theorem proves these shapes for first-departure drifts opposite to subgradients of $\|\cdot\|_\infty$ and $\|\cdot\|_1$. For related continuum growth models, Dembo and Yang [2025a,b] obtained limits described by geometric flow and regularized Kardar–Parisi–Zhang equations.

Internal diffusion-limited aggregation [Lawler, Bramson, and Griffeath, 1992] and rotor-router aggregation [Levine and Peres, 2009] have Euclidean ball limit shapes. The former has radius fluctuations of order at most $\log r$ in the plane and $(\log r)^{1/2}$ in higher dimensions [Jerison et al., 2012, 2013; Asselah and Gaudillière, 2013a,b], and its rescaled fluctuations approach a variant of the Gaussian free field [Jerison, Levine, and Sheffield, 2014]. For the centrally excited walk, Theorem 1.3 gives much larger planar radius and width fluctuations; the fixed-site covariance in Theorem 8.3 is again expressed through the Green function. See Levine and Peres [2017] for aggregation models.

For Eulerian walkers [Priezzhev, Dhar, Dhar, and Krishnamurthy, 1996], simulations suggested a circular planar range of radius of order $t^{1/3}$ [Kapri and Dhar, 2009]. Florescu, Levine, and Peres [2016] proved that every simple rotor walk on $\mathbb{Z}^d$ visits at least $cn^{d/(d+1)}$ sites in n steps. For clockwise planar rotors with independent uniform initial states, Bou-Rabee and Peres [2026a] proved convergence of the rescaled range to a deterministic compact convex set; whether it is a disk remains open.

1.6. Notation and conventions.

- The Euclidean norm is $|x|$, and $\|x\|_p$ is the ℓ^p norm for $1 \leq p \leq \infty$. The cardinality of a set $A \subset \mathbb{Z}^d$ is $|A|$, and the Lebesgue measure of a set $D \subset \mathbb{R}^d$ is $|D|$. The symmetric difference is Δ , and $t_+ := \max\{t, 0\}$ and $s \vee t := \max\{s, t\}$.
- Let $u_x := x/|x|$ for $x \in \mathbb{R}^d \setminus \{0\}$ and $u_0 := 0$, and let $\xi(0) := 0$.
- The letters $c > 0$ and $C > 0$ denote constants depending only on d , ε and the exponent p of the failure probability under discussion, and in Sections 2–5 also on the norm Ψ , but not on the choice of the subgradients $\xi(x)$; they may change from line to line. Every further dependence is stated, and a subscript on O , as in O_y , means that the implied constant also depends on y . We write $a_n \asymp b_n$ if $c \leq a_n/b_n \leq C$, and $O_{\mathbb{P}}(a_n)$ for random variables Y_n such that Y_n/a_n is tight.
- The σ -field generated by $X_0, \dots, X_j$ is $\mathcal{F}_j$. For martingales Z_1 and Z_2 , the brackets $\langle Z_1 \rangle_n$ and $\langle Z_1, Z_2 \rangle_n$ are the predictable quadratic variation and covariation through time n .
- The indicator that the step at time j is a first departure is $I_j := \mathbf{1}_{\{X_j \notin A_j\}}$, so that $\mathbb{E}(X_{j+1} - X_j \mid \mathcal{F}_j) = -\varepsilon I_j \xi(X_j)$, where $\xi(x) = u_x$ for the Euclidean norm.
- The function $\tilde{\ell}_n$ on $\mathbb{R}^d$ equals $\ell_n(x)$ on C_x for every $x \in \mathbb{Z}^d$; it vanishes off D_n .
- All estimates concern integers $n \geq 2$, so that $\log n > 0$. In a statement that holds with probability at least $1 - Cn^{-p}$ for all sufficiently large n , enlarging C makes it hold for every $n \geq 2$.

Use of artificial intelligence. The authors developed the main ideas of this work and directed its mathematical development. Throughout the project, the authors used large language models as research assistants: to explore specific questions arising in the proofs, to check the arguments

and the cited results, and to audit drafts of the paper. The exposition, the simulations and the figures were produced with AI agents, primarily Claude (Anthropic), under the close supervision of the authors. The results of this paper have been formalized in Lean 4. The development covers every theorem, proposition and lemma of the paper, including the limit-shape theorem for walks with a norm, and assumes no cited result as a hypothesis: it also proves the martingale central limit theorem and Stout's law of the iterated logarithm, in the forms the paper uses. It states its exact coverage and every difference from this paper at <https://github.com/nitromannitol/Centrally-Excited-Random-Walk>. The authors take responsibility for the mathematical content and presentation of the paper.

Concurrent work. A version of this manuscript containing our limit-shape and fluctuation results was recorded in our then-private [GitHub repository](#) on 30 September 2026, before the independent preprint of Nguyen [2026] appeared on 6 October 2026 (<https://arxiv.org/abs/2610.07724>). Nguyen proves convergence in probability, in Hausdorff distance, to an explicit ℓ^1 ball in every dimension $d \geq 2$. For his bias parameter $0 < \theta < 1$, the first-departure probabilities agree with ours for the ℓ^1 norm, with $\varepsilon = \theta/d$, away from the coordinate hyperplanes; the probabilities differ on these hyperplanes, and the limiting balls coincide. For our walk, we prove almost-sure shape and uniform local-time convergence for every norm, together with quantitative fluctuation bounds.

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2. DRIFTS GIVEN BY A NORM

In this section we give the precise setting for a drift opposite to a subgradient of a norm and state the limit shape theorem in this setting, Theorem 2.1, of which Theorem 1.1 is the Euclidean case. We also prove the identity for the potential of a ball of the norm on which its proof rests (Lemma 2.2). Sections 3–5 work in this setting and prove Theorem 2.1 at the end of Section 5.

Let Ψ be a norm on $\mathbb{R}^d$, let B_Ψ be its unit ball, and let $\partial\Psi(x)$ be the set of subgradients of Ψ at x , as in Section 1.3; if Ψ is differentiable at x , then $\partial\Psi(x) = \{\nabla\Psi(x)\}$. At each site $x \neq 0$ fix $\xi(x) \in \partial\Psi(x)$, with coordinates $\xi_i(x)$. The **centrally excited random walk with norm Ψ** moves like simple random walk, except that on its first departure from each site $x \neq 0$ it moves to $x \pm e_i$ with probability

$$(2.1) \quad \frac{1}{2d} \mp \frac{\varepsilon}{2} \xi_i(x) \quad \text{for } 1 \leq i \leq d.$$

We assume that

$$(2.2) \quad \varepsilon \max_{1 \leq i \leq d} \Psi(e_i) < \frac{1}{d}.$$

Every $\xi \in \partial\Psi(x)$ satisfies $\xi \cdot y \leq \Psi(y)$ for every y , with equality at $y = x$ (Euler's relation); so $|\xi_i(x)| \leq \Psi(e_i)$, and (2.2) makes the probabilities (2.1) positive. The step at the first departure from x has conditional mean $-\varepsilon\xi(x)$, whose inner product with x is $-\varepsilon\Psi(x) < 0$. The Euclidean norm, with $\xi(x) = x/|x|$, gives the walk of Section 1, and then (2.2) reads $\varepsilon < 1/d$. Let

$$(2.3) \quad r_n := \left(\frac{(d+1)n}{2d\varepsilon|B_\Psi|} \right)^{1/(d+1)},$$

which is (1.1) when Ψ is the Euclidean norm; the cone $2d\varepsilon(r_n - \Psi(v))_+$ has integral n over $\mathbb{R}^d$.

In Sections 2–5, Ψ is a norm, $\varepsilon > 0$ satisfies (2.2), r_n is given by (2.3), and U_D is the potential (2.4) defined below; at a point $v \in \mathbb{R}^d$, $\nabla\Psi(v)$ is the gradient, which exists for almost every v . Let $\Lambda_\Psi := \max_{|u|=1} \Psi(u)$ and $c_\Psi := \min_{|u|=1} \Psi(u)$; every $\xi \in \partial\Psi(x)$ satisfies $\xi \cdot x = \Psi(x)$ and $|\xi| \leq \Lambda_\Psi$. In Sections 6–9, Ψ is the Euclidean norm, so that $\xi(x) = x/|x|$, r_n is given by (1.1), and U_D is the potential (1.9).

Theorem 2.1 (Limit shape for a norm). *Let $d \geq 2$, let Ψ be a norm on $\mathbb{R}^d$, let $\xi(x) \in \partial\Psi(x)$ be chosen at the sites $x \in \mathbb{Z}^d \setminus \{0\}$, let $\varepsilon > 0$ satisfy (2.2), and let r_n be as in (2.3). Then, almost surely, the centrally excited random walk with norm Ψ has the following properties.*

(i) Shape: for every $0 < \eta < 1$ and all sufficiently large n ,

$$\{x \in \mathbb{Z}^d : \Psi(x) < (1 - \eta)r_n\} \subset A_n \subset \{x \in \mathbb{Z}^d : \Psi(x) < (1 + \eta)r_n\}.$$

(ii) Local times: as $n \rightarrow \infty$,

$$\frac{1}{r_n} \max_{x \in \mathbb{Z}^d} |\ell_n(x) - 2d\varepsilon(r_n - \Psi(x))_+| \rightarrow 0.$$

(iii) Recurrence: the walk visits every site of $\mathbb{Z}^d$ infinitely often.

Proposition 5.1 gives rates: for every $p > 0$ there is a constant $C(d, \varepsilon, \Psi, p)$, not depending on the choice of ξ , such that with probability at least $1 - Cn^{-p}$ the maximum in part (ii) is at most $Cr_n^{5/6}(\log n)^{1/6}$ if $d = 2$ and $Cr_n^{d/(d+1)}(\log n)^{1/(d+1)}$ if $d \geq 3$, and the inclusions of part (i) hold with ηr_n replaced by $\log n$ times this bound on the maximum. Part (i) gives $|A_n|/r_n^d \rightarrow |B_\Psi|$, and for the Euclidean norm Theorem 1.2 gives smaller error bounds.

2.1. The potential of a norm ball. For a norm Ψ , the potential (1.9) is defined with the gradient of Ψ in place of $v/|v|$: for a bounded measurable set $D \subset \mathbb{R}^d$ and $y \in \mathbb{R}^d$, let

$$(2.4) \quad U_D(y) := \frac{2\varepsilon}{\omega_d} \int_D \nabla \Psi(v) \cdot \frac{v - y}{|v - y|^d} dv$$

be the potential generated by the vector field $-\varepsilon \nabla \Psi$ on D ; for the Euclidean norm it is (1.9).

For every Lipschitz function f with compact support and every $y \in \mathbb{R}^d$,

$$(2.5) \quad \int_{\mathbb{R}^d} \nabla f(v) \cdot \frac{v - y}{|v - y|^d} dv = -d\omega_d f(y).$$

To prove it, write $v = y + tu$ with $t > 0$ and $|u| = 1$, so that $(v - y)|v - y|^{-d} dv = u dt dS(u)$. By Rademacher's and Fubini's theorems, for almost every u the Lipschitz function $t \mapsto f(y + tu)$ has derivative $\nabla f(y + tu) \cdot u$ for almost every $t > 0$; this function vanishes for large t , so its derivative integrates to $-f(y)$ over $t > 0$, and the unit sphere has area $d\omega_d$.

Lemma 2.2 (The potential of a norm ball). *Let Ψ be a norm, let $\rho > 0$, and let $y \in \mathbb{R}^d$. Then*

$$(2.6) \quad U_{\{\Psi < \rho\}}(y) = 2d\varepsilon(\rho - \Psi(y))_+.$$

Proof. The set $\{\Psi < \rho\}$ is bounded, and $\{\Psi = \rho\}$ has measure zero because Ψ is homogeneous, so $\nabla(\rho - \Psi)_+ = -\mathbf{1}_{\{\Psi < \rho\}} \nabla \Psi$ almost everywhere. Applying (2.5) to the Lipschitz function $f = (\rho - \Psi)_+$, which has compact support, proves the lemma. $\square$

For the Euclidean norm, Lemma 2.2 is the identity (1.10).

3. LOCAL TIMES AND THE POTENTIAL

In this section we prove Lemma 3.1, which bounds the local times on every time interval by the number of first departures in it and compares the local times with the potential U_{D_n} . Dynkin's formula expresses each local time as U_{D_n} minus a martingale, up to an error term. We bound the bracket of this martingale by a weighted sum of the local times, and use convexity to bound the error from replacing the subgradients $\xi(x)$ chosen at the sites by the gradient $\nabla \Psi$ on the cells (Lemma 3.3).

For integers $0 \leq s < t \leq n$, let

$$M_{s,t} := \max_{x \in \mathbb{Z}^d} \sum_{s \leq j < t} \mathbf{1}_{\{X_j = x\}} \quad \text{and} \quad k_{s,t} := \sum_{s \leq j < t} I_j.$$

Lemma 3.1. *Let $p > 0$. There exists $C(d, \varepsilon, \Psi, p) < \infty$ such that, for every integer $n \geq 2$, the following estimates hold simultaneously with probability at least $1 - Cn^{-p}$.*

(i) Largest local time:

$$(3.1) \quad \max_{x \in \mathbb{Z}^d} \ell_n(x) \leq C|A_n|^{1/d} + C(\log n)^2.$$

(ii) Local times on time intervals: for all integers $0 \leq s < t \leq n$,

$$(3.2) \quad M_{s,t} \leq Ck_{s,t}^{1/d} + C(\log n)^2.$$

(iii) Approximation by the potential: for every $y \in \mathbb{R}^d$ with $|y| \leq 2n$,

$$(3.3) \quad |\tilde{\ell}_n(y) - U_{D_n}(y)| \leq C \log n + C \begin{cases} (\max_x \ell_n(x))^{1/2} \log n, & d = 2, \\ (\max_x \ell_n(x) \log n)^{1/2}, & d \geq 3 \end{cases}.$$

Once $\max_x \ell_n(x) \leq Cr_n$, as on the event of Proposition 4.1, the right side of (3.3) is at most $C\sqrt{r_n} \log n = o(r_n)$.

3.1. Dynkin's formula for the local times. Let $\Delta f(x) := (2d)^{-1} \sum_{|e|=1} (f(x+e) - f(x))$, summed over the unit vectors $e \in \mathbb{Z}^d$, be the discrete Laplacian. For $d = 2$, let g be the potential kernel of simple random walk, normalized by $g(0) = 0$. For $d \geq 3$, let $g := -G$, where the Green function $G(x)$ is the expected number of visits to x by simple random walk from the origin. In both cases $\Delta g = \mathbf{1}_{\{0\}}$; for $d = 2$ see Lawler and Limic [2010, Proposition 4.4.2]. By Lawler and Limic [2010, Theorems 4.3.1 and 4.4.4], as $|x| \rightarrow \infty$,

$$(3.4) \quad g(x) = \begin{cases} \frac{2}{\pi} \log |x| + c_g + O(|x|^{-2}), & d = 2, \\ -\frac{2}{(d-2)\omega_d} |x|^{2-d} + O(|x|^{-d}), & d \geq 3 \end{cases},$$

where c_g is a constant. Let $\bar{\nabla} g(x)$ be the vector with coordinates $(g(x+e_i) - g(x-e_i))/2$ for $1 \leq i \leq d$. Then

$$(3.5) \quad \bar{\nabla} g(x) = \frac{2}{\omega_d} \frac{x}{|x|^d} + O(|x|^{-d}) \quad \text{and} \quad |g(x+e) - g(x)| \leq C(1 + |x|)^{1-d},$$

the first for $x \neq 0$, the second for every $x \in \mathbb{Z}^d$ and every unit vector $e \in \mathbb{Z}^d$; see Lawler and Limic [2010, Corollaries 4.3.3 and 4.4.5].

For $f: \mathbb{Z}^d \rightarrow \mathbb{R}$ with bounded nearest-neighbor increments, let

$$\mathcal{M}_t^f := f(X_t) - f(X_0) - \sum_{j < t} \mathbb{E}(f(X_{j+1}) - f(X_j) \mid \mathcal{F}_j)$$

be the martingale in Dynkin's formula for f . By (2.1), the conditional expectation in the sum is $\Delta f(X_j) - \varepsilon I_j \xi(X_j) \cdot \bar{\nabla} f(X_j)$, with $\bar{\nabla} f$ defined like $\bar{\nabla} g$. For $y \in \mathbb{Z}^d$ write $g_y := g(\cdot - y)$ and $\mathcal{M}^y := \mathcal{M}^{g_y}$. Since $\Delta g_y = \mathbf{1}_{\{y\}}$ and each site of A_n has its first departure before time n ,

$$(3.6) \quad \ell_n(y) = g(X_n - y) - g(-y) + \varepsilon \sum_{x \in A_n} \xi(x) \cdot \bar{\nabla} g(x - y) - \mathcal{M}_n^y.$$

The difference $g(X_n - y) - g(-y)$ is $O(\log n)$ for $d = 2$ and $O(1)$ for $d \geq 3$, uniformly over $y \in \mathbb{Z}^d$ with $|y| \leq 3n$. By (3.5), the increments of $\mathcal{M}^y$ are bounded, and

$$(3.7) \quad \langle \mathcal{M}^y \rangle_n \leq CW_y, \quad \text{where} \quad W_y := \sum_{x \in \mathbb{Z}^d} \ell_n(x) (1 + |x - y|)^{2-2d}.$$

Lemma 3.2. *Let $p > 0$. There exists $C(p) < \infty$ such that the following holds for every integer $n \geq 2$ and every real martingale Z whose increments have absolute value at most 1: with probability at least $1 - Cn^{-p}$,*

$$(3.8) \quad |Z_n - Z_0| \leq C(\sqrt{\langle Z \rangle_n \log n} + \log n).$$

Proof. Apply Freedman [1975, Theorem 1.6] on the dyadic intervals for $\langle Z \rangle_n$, with thresholds $C(p)(\sqrt{2^i \log n} + \log n)$. Each interval has failure probability at most $2n^{-p-1}$. Since $\langle Z \rangle_n \leq n$, a union bound over $O(\log n)$ intervals proves (3.8). $\square$

We apply Lemma 3.2 to constant multiples of possibly stopped martingales, with union bounds over polynomially many deterministic sites and time intervals; the resulting estimates then hold also at sites and intervals that depend on the path.

Let μ be the distributional Laplacian of Ψ on $\mathbb{R}^d$, so that $\int \phi d\mu = \int \Psi(v) \sum_{i=1}^d \partial_i^2 \phi(v) dv$ for every smooth compactly supported function ϕ on $\mathbb{R}^d$. The distributional Laplacian μ is a positive Radon measure, because the smooth convex functions obtained by mollifying Ψ have nonnegative Laplacian. For $y \in \mathbb{R}^d$ and $R > 0$, let ϕ be smooth with $0 \leq \phi \leq 1$, $\phi = 1$ on $B(y, R)$, support in $B(y, 2R)$ and $|\nabla \phi| \leq 2/R$. Integrating by parts and using $|\nabla \Psi| \leq \Lambda_\Psi$ gives

$$(3.9) \quad \mu(B(y, R)) \leq \int \phi d\mu = - \int \nabla \Psi \cdot \nabla \phi \leq 2^{d+1} \omega_d \Lambda_\Psi R^{d-1}.$$

Lemma 3.3. *There exists $C(d) < \infty$ such that, for every $x \in \mathbb{Z}^d$,*

$$\int_{C_x} |\nabla \Psi(v) - \xi(x)| dv \leq C\mu(B(x, 6\sqrt{d})).$$

Proof. The vector $\xi(x)$ is a subgradient of Ψ at x ; for $x = 0$, where $\xi(0) = 0$, this holds because $\Psi \geq 0$. So $h(v) := \Psi(v) - \Psi(x) - \xi(x) \cdot (v - x)$ is convex and nonnegative, $h(x) = 0$, $\nabla h = \nabla \Psi - \xi(x)$ almost everywhere, and the distributional Laplacian of h is μ . At every point v where h is differentiable, convexity of $s \mapsto h(v + se_i)$ and $h \geq 0$ give $|\partial_i h(v)| \leq h(v + e_i) + h(v - e_i)$. The cells $C_{x \pm e_i}$ are disjoint and lie in $B(x, A)$, where $A := 2\sqrt{d}$, so integrating over C_x and summing over i gives

$$\int_{C_x} |\nabla h(v)| dv \leq \int_{B(x, A)} h(v) dv.$$

Let h_r be the convolution of h with a smooth mollifier supported in $B(0, r)$, where $0 < r < A$. For a unit vector u , the derivative of the convex function $s \mapsto h_r(x + su)$ is nondecreasing, so $h_r(x + su) \leq h_r(x) + s(u \cdot \nabla h_r(x + 2Au))$ for $0 \leq s \leq A$. The Laplacian of h_r is the convolution of μ with the mollifier, so by the divergence theorem the average of $u \cdot \nabla h_r(x + 2Au)$ over unit vectors u is at most $C\mu(B(x, 2A + r))$. Integrating in polar coordinates gives $\int_{B(x, A)} h_r \leq \omega_d A^d h_r(x) + C\mu(B(x, 3A))$. Letting $r \rightarrow 0$, and using that $h_r \rightarrow h$ locally uniformly and $h(x) = 0$, proves the lemma. $\square$

Proof of Lemma 3.1. Step 1: The martingales on time intervals. Let $y \in \mathbb{Z}^d$ with $|y| \leq 3n$. Every site visited before time n lies within distance $4n$ of y , and the weights $(1 + |x - y|)^{2-2d}$ of the sites x with $|x - y| \leq 4n$ sum to $O(\log n)$ for $d = 2$ and $O(1)$ for $d \geq 3$. So $\langle \mathcal{M}^y \rangle_t - \langle \mathcal{M}^y \rangle_s$ is at most $CM_{s,t}$ times the sum of these weights. Lemma 3.2 for the martingales $j \mapsto \mathcal{M}_{(s+j) \wedge t}^y - \mathcal{M}_s^y$, with a union bound over the $O(n^{d+2})$ pairs of an interval and a site y , gives an event of probability at least $1 - Cn^{-p}$ on which, for all integers $0 \leq s < t \leq n$ and all $y \in \mathbb{Z}^d$ with $|y| \leq 3n$,

$$(3.10) \quad |\mathcal{M}_t^y - \mathcal{M}_s^y| \leq C \log n + C \begin{cases} \sqrt{M_{s,t}} \log n, & d = 2, \\ \sqrt{M_{s,t}} \log n, & d \geq 3 \end{cases}.$$

We work on this event.

Step 2: Local times on time intervals. For every $y \in \mathbb{R}^d$ and every set of k lattice sites, the sum of $(1 + |x - y|)^{1-d}$ over the set is at most $Ck^{1/d}$, since it is largest for the k sites nearest to y .

Subtract (3.6) with s in place of n from (3.6) with t in place of n , so that the left side becomes the number of departures from y at the times $s \leq j < t$, and maximize over the sites y with $|y| \leq 3n$, which include all sites visited before time n . The differences of the terms $g(X_t - y)$ and $g(X_s - y)$ contribute $O(\log n)$, and (3.10) bounds the martingale terms. The difference of the sums over A_t and A_s runs over the $k_{s,t}$ sites of $A_t \setminus A_s$, so by the bound above and (3.5) it is at most $Ck_{s,t}^{1/d}$. Young's inequality absorbs the term with $\sqrt{M_{s,t}}$, which proves (3.2). Taking $s = 0$ and $t = n$ gives (3.1), since $M_{0,n} = \max_x \ell_n(x)$ and $k_{0,n} = |A_n|$.

Step 3: Comparison with the potential. Let $y \in \mathbb{Z}^d$ with $|y| \leq 3n$. By (3.6), the bound on $g(X_n - y) - g(-y)$ and (3.10) with $s = 0$ and $t = n$, the local time $\ell_n(y)$ differs from $\varepsilon \sum_{x \in A_n} \xi(x) \cdot \bar{\nabla} g(x - y)$ by at most the right side of (3.3). The summand at $x = y$ vanishes because $\bar{\nabla} g(0) = 0$. For $x \neq y$, replace $\bar{\nabla} g(x - y)$ by $(2/\omega_d)(x - y)/|x - y|^d$; by (3.5), the errors sum to at most $C \sum_{|x| \leq n} (1 + |x - y|)^{-d} \leq C \log n$. Replacing each kernel value by its average over the cell C_x also changes the sum by at most $C \log n$: for cells at distance at least one from y each error is $O((1 + |x - y|)^{-d})$, and the cells near y , including C_y if $y \in A_n$, contribute $O(1)$ because $|v - y|^{1-d}$ is locally integrable. The last replacement, of $\xi(x)$ by $\nabla \Psi(v)$ for $v \in C_x$, changes the sum by at most a constant times

$$\sum_{x \in A_n} \int_{C_x} |v - y|^{1-d} |\xi(x) - \nabla \Psi(v)| dv.$$

The terms with $|x - y| \leq 8\sqrt{d}$ contribute $O(1)$, because $|\xi(x)|$ and $|\nabla \Psi(v)|$ are at most Λ_Ψ and $|v - y|^{1-d}$ is locally integrable. If $|x - y| > 8\sqrt{d}$, then $|v - y| \geq |x - y|/2$ for $v \in C_x$, and $|x - y|/4 \leq |z - y| \leq 2|x - y|$ for $z \in B(x, 6\sqrt{d})$; so Lemma 3.3 bounds the summand of x in the last display by $C \int_{B(x, 6\sqrt{d})} |z - y|^{1-d} d\mu(z)$. Each point of $\mathbb{R}^d$ lies in at most C of the balls $B(x, 6\sqrt{d})$ with $x \in \mathbb{Z}^d$, and those with $|x| \leq n$ and $|x - y| > 8\sqrt{d}$ lie in $\{z : 1 \leq |z - y| \leq 8n\}$. So the terms with $|x - y| > 8\sqrt{d}$ contribute at most $C \int_{\{1 \leq |z - y| \leq 8n\}} |z - y|^{1-d} d\mu(z) \leq C \log n$, because (3.9) bounds the integral over each of the $O(\log n)$ annuli $\{2^k \leq |z - y| < 2^{k+1}\}$ by a constant. After the three replacements the sum $\varepsilon \sum_{x \in A_n} \xi(x) \cdot \bar{\nabla} g(x - y)$ has become $U_{D_n}(y)$, which proves (3.3) for $y \in \mathbb{Z}^d$ with $|y| \leq 3n$.

Finally, for $y \in \mathbb{R}^d$ with $|y| \leq 2n$, let z be the site with $y \in C_z$. Then $\tilde{\ell}_n(y) = \ell_n(z)$, and splitting the integral (2.4) at $|v - z| = 2\sqrt{d}$, as for the cell averages, gives $|U_{D_n}(y) - U_{D_n}(z)| \leq C \log n$. This proves (3.3). $\square$

For every $p > 0$, Lemma 3.2 and (3.7) give, with probability at least $1 - Cn^{-p}$, simultaneously for all $y \in \mathbb{Z}^d$ with $|y| \leq 3n$,

$$(3.11) \quad |\mathcal{M}_n^y| \leq C(\sqrt{W_y \log n} + \log n).$$

The deterministic replacements in Step 3 of the proof of Lemma 3.1 give, for every $y \in \mathbb{Z}^d$ with $|y| \leq 3n$,

$$(3.12) \quad \ell_n(y) = U_{D_n}(y) - \mathcal{M}_n^y + O(\log n),$$

and splitting the integrals defining $U_{D_n}(y)$ and $U_{D_n}(z)$ at distance $2\sqrt{d}$ from y and z , as in that proof, gives, for all $y, z \in \mathbb{R}^d$ with $|y - z| \leq \sqrt{d}$ and $|y|, |z| \leq 3n$,

$$(3.13) \quad |U_{D_n}(y) - U_{D_n}(z)| \leq C \log n.$$

3.2. Bounds on the potential. The potential of a set is bounded by its volume, and by the width of a shell of Ψ that contains it. Let $D \subset \mathbb{R}^d$ be a bounded measurable set. For every $y \in \mathbb{R}^d$, the integral of $|v - y|^{1-d}$ over D is at most its integral over the ball of volume $|D|$ centered at y , which

is $C|D|^{1/d}$; since $|\nabla\Psi| \leq \Lambda_\Psi$, there exists $C(d, \Psi) < \infty$ such that

$$(3.14) \quad \sup_{y \in \mathbb{R}^d} |U_D(y)| \leq C\varepsilon|D|^{1/d}.$$

The function U_D is continuous, because it is the convolution of the bounded, compactly supported vector field $\mathbf{1}_D \nabla \Psi$ with a kernel that is the sum of an integrable function and a bounded one. We write dS for surface measure on a sphere.

Convexity of Ψ along lines bounds the potential of a set that lies between two balls of Ψ about the origin.

Lemma 3.4. *Let $b \geq 0$ and $a > 0$, and let D be a measurable subset of $\{v \in \mathbb{R}^d : b \leq \Psi(v) \leq b + a\}$. Then*

$$\sup_{y \in \mathbb{R}^d} |U_D(y)| \leq 2d\varepsilon a.$$

Proof. Fix $y \in \mathbb{R}^d$, write $v = y + tu$ with $t > 0$ and $|u| = 1$, and let $h_u(t) := \Psi(y + tu)$. As in the proof of (2.5), for almost every u the function h_u has derivative $\nabla\Psi(y + tu) \cdot u$ for almost every $t > 0$, so

$$U_D(y) = \frac{2\varepsilon}{\omega_d} \int_{\{|u|=1\}} \int_0^\infty \mathbf{1}_D(y + tu) h'_u(t) dt dS(u).$$

Since h_u is convex, there is $t_0 \in [0, \infty]$ such that h_u is nonincreasing on $[0, t_0]$ and nondecreasing on $[t_0, \infty)$. On each of these intervals, the change of variables $s = h_u(t)$ bounds the integral of $\mathbf{1}_{[b, b+a]}(h_u(t))|h'_u(t)|$ by a . Since $0 \leq \mathbf{1}_D(y + tu) \leq \mathbf{1}_{[b, b+a]}(h_u(t))$, and $h'_u \leq 0$ on the first interval and $h'_u \geq 0$ on the second, the inner integral lies in $[-a, a]$. Since the unit sphere has area $d\omega_d$, it follows that $|U_D(y)| \leq (2\varepsilon/\omega_d) d\omega_d a = 2d\varepsilon a$, which proves the lemma; by Lemma 2.2, equality holds at $y = 0$ when $D = \{b \leq \Psi < b + a\}$. $\square$

4. COARSE BOUNDS ON THE RANGE AND THE LOCAL TIMES

We prove that, with high probability, $|A_n|$ has order r_n^d , and $R_{\text{out}}(n) = \max_{j \leq n} |X_j|$ and the largest local time have order r_n (Proposition 4.1). The drift at first departures opposes the crossing of a region with few visited sites, and (3.2) bounds the time spent there. The crossing argument of Section 4.1, which rests on these two facts, bounds $R_{\text{out}}(n)$, for every $a \geq 0$, by a/c_Ψ plus $C \log n$ times one plus the largest local time at the sites x with $\Psi(x) \geq a$ (inequality (4.7)). By Lemma 3.1, the largest local time is of order at most $|A_n|^{1/d}$, so $R_{\text{out}}(n) \leq C|A_n|^{1/d} \log n$. The potential at a point where the boundary of the largest ball of Ψ in D_n meets the closure of a cell disjoint from D_n (Section 4.2) then shows that the local times outside this ball are much smaller than $|A_n|^{1/d} / \log n$, so that $R_{\text{out}}(n) \leq C|A_n|^{1/d}$, and that the local times are close to a cone, whose integral bounds $|A_n|$.

Proposition 4.1. *Let $p > 0$. There exist $c(d, \varepsilon, \Psi, p) > 0$ and $C(d, \varepsilon, \Psi, p) < \infty$ such that, for every integer $n \geq 2$, with probability at least $1 - Cn^{-p}$,*

$$cr_n^d \leq |A_n| \leq Cr_n^d, \quad cr_n \leq \max_{x \in \mathbb{Z}^d} \ell_n(x) \leq Cr_n, \quad cr_n \leq R_{\text{out}}(n) \leq Cr_n.$$

4.1. Crossing a region with few visited sites. A crossing against the drift requires a large increment of the martingale

$$(4.1) \quad Z_t := X_t + \varepsilon \sum_{j < t} I_j \xi(X_j),$$

which takes values in $\mathbb{R}^d$ and has bounded increments. Azuma's inequality in each coordinate and a union bound over the pairs $s < t$ give, for every $p > 0$, with probability at least $1 - Cn^{-p}$,

$$(4.2) \quad |Z_t - Z_s| \leq C\sqrt{(t-s) \log n} \quad \text{for } 0 \leq s < t \leq n.$$

Lemma 4.2. *On the event in (4.2), with the constant C of (4.2), the following holds for every unit vector u and all integers $0 \leq s < t \leq n$. If $u \cdot \xi(X_j) \geq 0$ at every first departure time j with $s \leq j < t$, then*

$$u \cdot (X_t - X_s) \leq C\sqrt{(t-s)\log n}.$$

Proof. By (4.1), $u \cdot (X_t - X_s) = u \cdot (Z_t - Z_s) - \varepsilon \sum_{s \leq j < t} I_j u \cdot \xi(X_j) \leq |Z_t - Z_s|$, and (4.2) bounds the right side. $\square$

The martingales of the functions $(q \cdot x - a)_+$ bound the number of sites $x \in A_n$ at which $q \cdot x$ exceeds a given level (Step 2 of the proof of Lemma 4.3); we bound them simultaneously for all vectors q in a finite deterministic set, because q will depend on the path. For $q \in \mathbb{R}^d$ and $a \in \mathbb{R}$, let $\mathcal{M}^{q,a}$ be the martingale $\mathcal{M}^f$ of Section 3.1 with $f(x) = (q \cdot x - a)_+$. If $|q| \leq \Lambda_\Psi$, then one step changes $q \cdot X_j$ by at most Λ_Ψ , so the increments of $\mathcal{M}^{q,a}$ are at most C , and the bracket of $\mathcal{M}^{q,a}$ through time n is at most $C \sum_{x: q \cdot x > a - \Lambda_\Psi} \ell_n(x)$, because f is constant on x and its neighbors unless $q \cdot x > a - \Lambda_\Psi$. Fix $p > 0$. Lemma 3.2 and a union bound over the $O(n^{d+1})$ pairs (q, a) with $q = \Lambda_\Psi u_x$ for some $x \in \mathbb{Z}^d$ with $0 < |x| \leq n$ and $a \in \Lambda_\Psi \mathbb{Z} \cap [0, \Lambda_\Psi n]$ give, with probability at least $1 - Cn^{-p}$, simultaneously for all these pairs,

$$(4.3) \quad |\mathcal{M}_n^{q,a}| \leq C \left(\sqrt{\log n \sum_{x: q \cdot x > a - \Lambda_\Psi} \ell_n(x)} + \log n \right).$$

Lemma 4.3. *Let $\alpha > 0$. There exists $C(d, \varepsilon, \Psi, \alpha, p) < \infty$ with the following property. Let $q \in \mathbb{R}^d$ satisfy $|q| \leq \Lambda_\Psi$, and consider a path of the walk that satisfies (4.2), and (4.3) for this q and every $a \in \Lambda_\Psi \mathbb{Z} \cap [0, \Lambda_\Psi n]$. Let x_* be one of the sites $X_0, \dots, X_n$ with $q \cdot X_j \leq T := q \cdot x_*$ for $0 \leq j \leq n$, and let $h > 0$ be such that $q \cdot \xi(X_j) \geq \alpha$ for every $0 \leq j \leq n$ with $q \cdot X_j > T - h$. For $b \geq 0$ let $L_b := 1 + \max\{\ell_n(x) : x \in \mathbb{Z}^d, q \cdot x \geq b\}$. Then every $b \geq 0$ satisfies*

$$(4.4) \quad T \leq \max\{T - h, b\} + CL_b \log n.$$

Proof. Fix $b \geq 0$. Steps 1–3 show that few sites of A_n lie where $q \cdot x$ exceeds a level slightly above $\max\{T - h, b\}$, and Step 4 shows that the walk cannot reach x_* through so few sites.

Step 1: Local times and drift beyond a level. For $a \in \mathbb{R}$ let $\mathcal{N}(a)$ be the number of sites $x \in A_n$ with $q \cdot x > a$. Let a_0 be the least element of $\Lambda_\Psi \mathbb{Z}$ that is at least $\max\{T - h, b + \Lambda_\Psi\}$. If $a \geq a_0$, then:

- (a) every $x \in A_n$ with $q \cdot x > a - \Lambda_\Psi$ satisfies $q \cdot x > b$, so $\ell_n(x) < L_b$;
- (b) every $x \in A_n$ with $q \cdot x > a$ satisfies $q \cdot x > T - h$, so $q \cdot \xi(x) \geq \alpha$ by hypothesis.

Step 2: Comparison of neighboring levels. Since $|q| \leq \Lambda_\Psi$ and $|x_*| \leq n$, we have $T \leq \Lambda_\Psi n$, and $\mathcal{N}(a) = 0$ for $a \geq T$; so it suffices to consider levels $a \leq \Lambda_\Psi n$. Fix $a \in \Lambda_\Psi \mathbb{Z}$ with $a_0 \leq a \leq \Lambda_\Psi n$, and put $\phi(x) := (q \cdot x - a)_+$. Dynkin's formula gives

$$\phi(X_n) = \sum_{j < n} [\Delta \phi(X_j) - \varepsilon I_j \xi(X_j) \cdot \bar{\nabla} \phi(X_j)] + \mathcal{M}_n^{q,a}.$$

If $q \cdot x \geq a + \Lambda_\Psi$, then ϕ is linear on x and its neighbors, because $|q| \leq \Lambda_\Psi$, so $\Delta \phi(x) = 0$ and $\bar{\nabla} \phi(x) = q$, and by (b) the drift term at the first departure from x is at most $-\varepsilon \alpha$; at later departures from x the drift term vanishes. If $q \cdot x \leq a - \Lambda_\Psi$, then ϕ vanishes at x and its neighbors. At the remaining sites, $|\Delta \phi|$ and $|\bar{\nabla} \phi|$ are at most C . Since $\phi(X_n) \geq 0$ and $\phi(X_0) = 0$, properties (a) and (b) and (4.3) give

$$\varepsilon \alpha \mathcal{N}(a + \Lambda_\Psi) \leq CL_b |\{x \in A_n : |q \cdot x - a| < \Lambda_\Psi\}| + C \sqrt{L_b \mathcal{N}(a - \Lambda_\Psi) \log n} + C \log n.$$

Hence, for every $a \in \Lambda_\Psi \mathbb{Z}$ with $a \geq a_0$,

$$(4.5) \quad \mathcal{N}(a + \Lambda_\Psi) \leq CL_b [\mathcal{N}(a - \Lambda_\Psi) - \mathcal{N}(a + \Lambda_\Psi)] + C \sqrt{L_b \mathcal{N}(a - \Lambda_\Psi) \log n} + C \log n.$$

Step 3: Iteration over the levels. Set $N_k := \mathcal{N}(a_0 - \Lambda_\Psi + 2k\Lambda_\Psi)$. Young's inequality in (4.5) gives $N_{k+1} \leq (1 - c/L_b)N_k + C \log n$. Since $N_0 \leq n$, iteration to $k = \lceil L_b \log n / c \rceil$ gives, with $\bar{a} := a_0 - \Lambda_\Psi + 2k\Lambda_\Psi$,

$$(4.6) \quad a_0 \leq \bar{a} \leq a_0 + CL_b \log n \quad \text{and} \quad \mathcal{N}(\bar{a}) \leq CL_b \log n.$$

Step 4: The bound on T . If $T \leq \bar{a} + \Lambda_\Psi$, then $T - a_0 \leq CL_b \log n$ by (4.6). Otherwise, $q \neq 0$ because $T > 0$; let σ be the first $j \leq n$ with $X_j = x_*$, and let $s := 1 + \max\{j < \sigma : q \cdot X_j \leq \bar{a}\}$. The set $\{j < \sigma : q \cdot X_j \leq \bar{a}\}$ contains 0 because $q \cdot X_0 = 0 \leq \bar{a}$, and $s < \sigma$ because $T > \bar{a} + \Lambda_\Psi$ and each step changes $q \cdot X_j$ by at most Λ_Ψ . For $s \leq j < \sigma$, the site X_j belongs to A_n and satisfies $q \cdot X_j > \bar{a} \geq a_0$, so (b) gives $q \cdot \xi(X_j) > 0$. By (4.6) and (a), at most $CL_b \log n$ distinct sites occur among $X_s, \dots, X_{\sigma-1}$, each with local time less than L_b , so $\sigma - s \leq CL_b^2 \log n$. Lemma 4.2 with $u = q/|q|$ and $t = \sigma$ gives $q \cdot (X_\sigma - X_s) \leq C \sqrt{(\sigma - s) \log n} \leq CL_b \log n$. Since $q \cdot X_\sigma = T$ and $q \cdot X_s \leq \bar{a} + \Lambda_\Psi$, the bound $T - a_0 \leq CL_b \log n$ holds in this case as well. As $a_0 \leq \max\{T - h, b + \Lambda_\Psi\} + \Lambda_\Psi$ and $L_b \geq 1$, this proves (4.4). $\square$

Let x_* be one of the sites $X_0, \dots, X_n$ with $|x_*| = R_{\text{out}}(n)$, so that $x_* \neq 0$ and $|x_*| \leq n$. Let $q := \Lambda_\Psi u_{x_*}$ and $T := q \cdot x_* = \Lambda_\Psi R_{\text{out}}(n)$, so that $q \cdot X_j \leq \Lambda_\Psi |X_j| \leq T$ for $0 \leq j \leq n$, and let $\theta := c_\Psi^2 / (8\Lambda_\Psi^2)$. If $q \cdot X_j > (1 - \theta)T$, then $X_j \neq 0$ and $u_{x_*} \cdot u_{X_j} > (1 - \theta)R_{\text{out}}(n)/|X_j| \geq 1 - \theta$, so $|u_{x_*} - u_{X_j}| < (2\theta)^{1/2}$, and Euler's relation and $|\xi(X_j)| \leq \Lambda_\Psi$ give

$$q \cdot \xi(X_j) \geq \Lambda_\Psi (u_{X_j} \cdot \xi(X_j) - \Lambda_\Psi |u_{x_*} - u_{X_j}|) \geq \Lambda_\Psi (c_\Psi - \Lambda_\Psi (2\theta)^{1/2}) = \frac{\Lambda_\Psi c_\Psi}{2}.$$

Let $a \geq 0$. Every $x \in \mathbb{R}^d$ with $q \cdot x \geq \Lambda_\Psi a / c_\Psi$ satisfies $\Psi(x) \geq c_\Psi |x| \geq a$. So, on the event on which (4.2) and (4.3) hold, Lemma 4.3 with $h = \theta T$, $\alpha = \Lambda_\Psi c_\Psi / 2$ and the level $\Lambda_\Psi a / c_\Psi$ in place of b gives

$$T \leq \max\left\{(1 - \theta)T, \frac{\Lambda_\Psi a}{c_\Psi}\right\} + C\left(1 + \max_{x \in \mathbb{Z}^d, \Psi(x) \geq a} \ell_n(x)\right) \log n.$$

If the maximum is $(1 - \theta)T$, then θT is at most the last term; if the maximum is $\Lambda_\Psi a / c_\Psi$, then $T - \Lambda_\Psi a / c_\Psi$ is at most the last term. Since $T = \Lambda_\Psi R_{\text{out}}(n)$, in both cases, for every $a \geq 0$,

$$(4.7) \quad R_{\text{out}}(n) \leq \frac{a}{c_\Psi} + C\left(1 + \max_{x \in \mathbb{Z}^d, \Psi(x) \geq a} \ell_n(x)\right) \log n.$$

For the Euclidean norm, $\xi(x) = u_x$ and $c_\Psi = 1$, and the maximum is over the sites with $|x| \geq a$.

4.2. The largest ball of the norm in the range. The following objects are used in Sections 4–7. Let $x \in \mathbb{R}^d$ and $\xi \in \partial\Psi(x)$, so that $\Psi(y) \geq \Psi(x) + \xi \cdot (y - x)$ for $y \in \mathbb{R}^d$. Euler's relation $\xi \cdot x = \Psi(x)$ turns this inequality into

$$(4.8) \quad \xi \cdot y \leq \Psi(y),$$

with equality when $y = x$. Let

$$b := \inf\{\Psi(y) : y \in \mathbb{R}^d \setminus D_n\} \quad \text{and} \quad E := D_n \setminus \{\Psi < b\},$$

so that $\{\Psi < b\} \subset D_n$. Choose $y_0 \in \overline{\mathbb{R}^d \setminus D_n}$ with $\Psi(y_0) = b$ and a cell C_z disjoint from D_n whose closure contains y_0 . Then $\ell_n(z) = 0$ and $\Psi(z) \geq b$. Let

$$H := U_{D_n}(y_0) \quad \text{and} \quad \mathcal{I} := \int_E (\Psi(v) - b) dv.$$

Lemma 2.2 gives $U_{\{\Psi < b\}}(y_0) = 0$, so $H = U_E(y_0)$. For almost every $v \in E$, the inequality (4.8) at v , with $\nabla\Psi(v)$ in place of ξ , gives

$$(4.9) \quad \nabla\Psi(v) \cdot (v - y_0) \geq \Psi(v) - b \geq 0.$$

Hence, if $E \subset B(y_0, \rho)$ for some $\rho > 0$, then

$$(4.10) \quad H \geq \frac{2\varepsilon}{\omega_d} \int_E \frac{\Psi(v) - b}{|v - y_0|^d} dv \geq \frac{c\mathcal{I}}{\rho^d}, \quad \text{so } \mathcal{I} \leq C\rho^d H.$$

On the event of Lemma 3.1, if $|y_0| \leq n$, then H is at most the right side of (3.3), because $\tilde{\ell}_n$ vanishes on C_z and U_{D_n} is continuous.

The integral $\mathcal{I}$ also bounds the potential of E . Let $a > 0$. The potential U_E is the sum of the potentials of the part of E where $\Psi \leq b + a$ and of the rest of E . By Lemma 3.4, the potential of the first part is at most $2d\varepsilon a$ in absolute value; the rest of E has volume at most $\mathcal{I}/a$, because $\Psi - b > a$ there, so (3.14) bounds the absolute value of the potential of the rest of E by $C(\mathcal{I}/a)^{1/d}$. If $\mathcal{I} > 0$, then the choice $a := \mathcal{I}^{1/(d+1)}$ gives

$$(4.11) \quad \sup_{y \in \mathbb{R}^d} |U_E(y)| \leq C\mathcal{I}^{1/(d+1)}.$$

If $\mathcal{I} = 0$, then $|E| = 0$, because $E \subset \{\Psi \geq b\}$ and $\{\Psi = b\}$ has measure zero, so (4.11) holds as well.

4.3. Proof of the coarse bounds.

Proof of Proposition 4.1. Work on the intersection of the events of Lemma 3.1 and of (4.2) and (4.3); its complement has probability at most Cn^{-p} . Let $s := |A_n|^{1/d}$, and take n large.

Step 1: First bounds. Since $n \leq |A_n| \max_x \ell_n(x)$, Lemma 3.1 gives

$$n \leq Cs^{d+1} + C(\log n)^2 s^d.$$

Hence $s \geq cr_n$, $\max_x \ell_n(x) \leq Cs$, and (4.7) with $a = 0$ gives $R_{\text{out}}(n) \leq Cs \log n$.

Step 2: The local times outside the largest ball. Let b , E , y_0 , H and $\mathcal{I}$ be as in Section 4.2. Since $\{\Psi < b\} \subset D_n$, we have $|B_\Psi|b^d \leq |D_n| = s^d$, so $b \leq Cs$ and $|y_0| \leq b/c_\Psi \leq Cs \leq n$. Since $\max_x \ell_n(x) \leq Cs$, the right side of (3.3) is at most $Cs^{1/2} \log n$, and hence, by the sentence after (4.10), which applies because $|y_0| \leq n$, so is H . Every point of D_n lies within distance $R_{\text{out}}(n) + \sqrt{d}$ of the origin, so $E \subset B(y_0, Cs \log n)$, and (4.10) gives

$$\mathcal{I} \leq C(s \log n)^d H \leq Cs^{d+1/2} (\log n)^{d+1}.$$

By Lemma 2.2, $U_{D_n} = 2d\varepsilon(b - \Psi)_+ + U_E$, and $\sup |U_E| \leq C\mathcal{I}^{1/(d+1)}$ by (4.11). With (3.3), these two facts give, for $|y| \leq 2n$,

$$(4.12) \quad |\tilde{\ell}_n(y) - 2d\varepsilon(b - \Psi(y))_+| \leq Cs^{1/2} \log n + C\mathcal{I}^{1/(d+1)} \leq Cs^{(2d+1)/(2d+2)} \log n,$$

and both $\tilde{\ell}_n(y)$ and $2d\varepsilon(b - \Psi(y))_+$ vanish for $|y| > 2n$. In particular, every $x \in \mathbb{Z}^d$ with $\Psi(x) \geq b$ satisfies $\ell_n(x) \leq Cs^{(2d+1)/(2d+2)} \log n$.

Step 3: The coarse bounds. By (4.7) with $a = b$ and Step 2,

$$R_{\text{out}}(n) \leq \frac{b}{c_\Psi} + Cs^{(2d+1)/(2d+2)} (\log n)^2,$$

and the last term is $o(s)$, because $s \geq cr_n$. Since $|A_n| \leq C(R_{\text{out}}(n) + 1)^d$, we have $s \leq C(R_{\text{out}}(n) + 1)$, so $b \geq cs$. The functions $\tilde{\ell}_n$ and $2d\varepsilon(b - \Psi)_+$ vanish outside D_n , whose volume is s^d , and their integrals are n and $2d\varepsilon|B_\Psi|b^{d+1}/(d+1)$; so (4.12) and $b \geq cs$ give

$$n \geq \frac{2d\varepsilon|B_\Psi|}{d+1} b^{d+1} - Cs^{d+(2d+1)/(2d+2)} \log n \geq cs^{d+1},$$

and (2.3) gives $|A_n| = s^d \leq Cr_n^d$. Then $\max_x \ell_n(x) \leq Cs \leq Cr_n$ and $R_{\text{out}}(n) \leq Cs \leq Cr_n$. The lower bound on $R_{\text{out}}(n)$ follows from $|A_n| \geq cr_n^d$ (Step 1) and $|A_n| \leq C(R_{\text{out}}(n) + 1)^d$, and the lower bound on $\max_x \ell_n(x)$ from $n \leq |A_n| \max_x \ell_n(x)$ and $|A_n| \leq Cr_n^d$. $\square$

5. THE LIMIT SHAPE

We prove Theorem 2.1 with rates (Proposition 5.1). Let $\{\Psi < b\}$ be the largest ball of Ψ about the origin contained in D_n , and let y_0 be a point of $\{\Psi = b\}$ in the closure of a cell disjoint from D_n (Section 4.2). By convexity, every point v of D_n outside the ball contributes at least $c(\Psi(v) - b)r_n^{-d}$ to $U_{D_n}(y_0)$, while Dynkin's formula at the center of this cell, a site outside A_n , now with the coarse bounds, makes $U_{D_n}(y_0)$ small (Lemma 5.2). So little of D_n lies outside the ball, and the quadratic martingale (5.5) shows that b is close to r_n . For the outer bound we use the crossing argument of Section 4.1 again: let x_* be one of the sites $X_0, \dots, X_n$ at which a linear function $x \mapsto q \cdot x$ is largest among these sites. If the first departure from each visited site where $q \cdot x$ is close to its maximum decreases $q \cdot x$ in conditional mean, then few sites of the range lie where $q \cdot x$ is large, and the walk cannot cross them to reach x_* (Lemma 4.3). When Ψ has corners, this hypothesis fails for the subgradient $q = \xi(x_*)$ at a site x_* where Ψ is largest: the drift $-\varepsilon\xi(y)$ at a visited site y where $q \cdot y$ is close to $q \cdot x_*$ can be orthogonal to $\xi(x_*)$. So we take for q the outer normal of the ball $\{\Psi \leq r_n\}$ at the point nearest to the visited site farthest from this ball in Euclidean distance.

Proposition 5.1. *Let Ψ be a norm, let $\xi(x) \in \partial\Psi(x)$ be subgradients chosen at the sites $x \in \mathbb{Z}^d \setminus \{0\}$, and let $\varepsilon > 0$ satisfy (2.2). For every $p > 0$ there exists $C(d, \varepsilon, \Psi, p) < \infty$, not depending on the choice of the subgradients, such that, for every integer $n \geq 2$, with probability at least $1 - Cn^{-p}$ the following hold simultaneously.*

(i) Radii:

$$(5.1) \quad \left| \inf_{y \notin D_n} \Psi(y) - r_n \right| \leq C \begin{cases} r_n^{3/4} (\log n)^{1/4}, & d = 2, \\ (r_n \log n)^{1/2}, & d \geq 3 \end{cases}$$

and

$$(5.2) \quad \max_{0 \leq j \leq n} \Psi(X_j) - r_n \leq C \begin{cases} r_n^{5/6} (\log n)^{7/6}, & d = 2, \\ r_n^{d/(d+1)} (\log n)^{(d+2)/(d+1)}, & d \geq 3 \end{cases}.$$

(ii) Local times:

$$(5.3) \quad \max_{x \in \mathbb{Z}^d} |\ell_n(x) - 2d\varepsilon(r_n - \Psi(x))_+| \leq C \begin{cases} r_n^{5/6} (\log n)^{1/6}, & d = 2, \\ r_n^{d/(d+1)} (\log n)^{1/(d+1)}, & d \geq 3 \end{cases}.$$

Throughout the rest of the paper let

$$(5.4) \quad q_n := \begin{cases} \sqrt{\frac{\log n}{r_n}}, & d = 2, \\ \frac{\log n}{r_n}, & d \geq 3 \end{cases}.$$

The bounds in (5.1), (5.2) and (5.3) are $C r_n q_n^{1/2}$, $C r_n q_n^{1/(d+1)} \log n$ and $C r_n q_n^{1/(d+1)}$.

5.1. Martingale estimates. We collect on one event the martingale bounds used in Sections 5–7. Fix $p > 0$. Since every step has squared length one and conditional mean $-\varepsilon I_j \xi(X_j)$ given $\mathcal{F}_j$, the process

$$\mathcal{Q}_t := \sum_{j < t} 2X_j \cdot (X_{j+1} - X_j + \varepsilon I_j \xi(X_j))$$

is a martingale, and Euler's relation gives

$$(5.5) \quad |X_n|^2 = n - 2\varepsilon \sum_{x \in A_n} \Psi(x) + \mathcal{Q}_n.$$

Azuma's inequality for $\mathcal{Q}$ stopped on leaving $B(0, k)$, followed by a union bound over $1 \leq k \leq n+1$, gives, with probability at least $1 - Cn^{-p}$,

$$(5.6) \quad |\mathcal{Q}_n| \leq C(R_{\text{out}}(n) + 1)\sqrt{n \log n}.$$

We also use the martingales $\mathcal{M}^{q,a}$ of Section 4.1 in the directions of Euclidean projections. For $x \in \mathbb{R}^d$, let $\Pi_n(x)$ be the point of the closed convex set $\{\Psi \leq r_n\}$ nearest to x in Euclidean distance. Lemma 3.2 and a union bound over the $O(n^{d+1})$ pairs (q, a) with $q = \Lambda_\Psi u_{x - \Pi_n(x)}$ for some $x \in \mathbb{Z}^d$ with $|x| \leq n$ and $\Psi(x) > r_n$, and $a \in \Lambda_\Psi \mathbb{Z} \cap [0, \Lambda_\Psi n]$, give (4.3) also for these pairs, with probability at least $1 - Cn^{-p}$.

Intersect the events of Proposition 4.1 and Lemma 3.1 with the events on which (4.2), (3.11) and (5.6) hold and on which (4.3) holds for the pairs of Section 4.1 and for those above; the complement has probability at most Cn^{-p} , and the arguments of Sections 5–7 are deterministic on this event. Let K be a constant with $R_{\text{out}}(n) \leq (K-1)r_n$ on the event of Proposition 4.1; then $D_n \subset B(0, Kr_n)$ for large n , and (5.6) gives

$$(5.7) \quad |\mathcal{Q}_n| \leq Cr_n \sqrt{n \log n},$$

where $\mathcal{Q}$ is the martingale of (5.5). By Proposition 4.1, $\max_x \ell_n(x) \leq Cr_n$, so (3.3) gives, for $y \in \mathbb{R}^d$ with $|y| \leq 2n$,

$$(5.8) \quad |\tilde{\ell}_n(y) - U_{D_n}(y)| \leq \delta, \quad \text{where } \delta := Cr_n^{1/2} \log n.$$

5.2. The inner radius and the local times. Let b, E, y_0, z, H and $\mathcal{I}$ be as in Section 4.2. Then $|v - y_0| \leq Cr_n$ for $v \in E$, because $D_n \subset B(0, Kr_n)$ and $\Psi(y_0) = b \leq Cr_n$, so (4.10) gives $\mathcal{I} \leq Cr_n^d H$. Since $\tilde{\ell}_n$ vanishes on C_z , the bound (5.8) gives $U_{D_n} \leq \delta$ on C_z , and hence $H \leq \delta$, because U_{D_n} is continuous and y_0 lies in the closure of C_z .

The potential H is small because the walk has not departed from z : up to $C \log n$, Dynkin's formula at z bounds H by $|\mathcal{M}_n^z|$, and the bracket of $\mathcal{M}^z$ is at most CW_z by (3.7). We bound W_z by the convolution of $\tilde{\ell}_n$ with a radial kernel at y_0 ; when $d = 2$ the bound $H \leq \delta$ then suffices, and when $d \geq 3$ we bound W_z by H itself. Both cases use the following identity. Let $s > 0$. For $d = 2$, the average of $\log |v - w|$ over the circle $\{|w| = s\}$ is $\log \max\{|v|, s\}$, and for $d \geq 3$, the average of $|v - w|^{2-d}$ over the sphere $\{|w| = s\}$ is $\max\{|v|, s\}^{2-d}$, by the mean-value property and radial harmonicity inside and outside the sphere. Differentiating in v , away from $|v| = s$, gives

$$(5.9) \quad \frac{1}{d\omega_d s^{d-1}} \int_{\{|w|=s\}} \frac{v-w}{|v-w|^d} dS(w) = |v|^{1-d} u_v \mathbf{1}_{\{|v|>s\}}.$$

Let χ be a nonnegative integrable radial function on $\mathbb{R}^d$, and let $m(s) := \int_{|\zeta|<s} \chi(\zeta) d\zeta$ for $s \geq 0$. Integrating (5.9) in polar coordinates gives, for almost every $y \in \mathbb{R}^d$, the following form of Newton's theorem [Lieb and Loss, 2001, Theorem 9.7]:

$$(5.10) \quad \int_{\mathbb{R}^d} \chi(\zeta) \frac{y-\zeta}{|y-\zeta|^d} d\zeta = m(|y|) \frac{y}{|y|^d}.$$

Let $D \subset \mathbb{R}^d$ be a bounded measurable set. By (5.10) with $v - y$ in place of y and the symmetry of χ , and by Fubini's theorem, which applies because χ is integrable and $\sup_w \int_D |v - w|^{1-d} dv < \infty$, every $y \in \mathbb{R}^d$ satisfies

$$(5.11) \quad (\chi * U_D)(y) = \frac{2\varepsilon}{\omega_d} \int_D \nabla \Psi(v) \cdot \frac{v-y}{|v-y|^d} m(|v-y|) dv.$$

For $D = E$ and $y = y_0$ the integrand is nonnegative by (4.9), and $0 \leq m \leq \|\chi\|_1$, so

$$(5.12) \quad (\chi * U_E)(y_0) \leq \|\chi\|_1 H.$$

Lemma 5.2. *There exists $C(d, \varepsilon, \Psi, p) < \infty$ such that, on the event of Section 5.1, for large n ,*

$$H \leq Cr_n q_n = C \begin{cases} \sqrt{r_n \log n}, & d = 2, \\ \log n, & d \geq 3 \end{cases}.$$

Proof. Since $\ell_n(z) = 0$, the bounds (3.12) and (3.11) at z , which apply because $|z| \leq Cr_n < 3n$, and (3.13) give

$$(5.13) \quad H \leq C(\sqrt{W_z \log n} + \log n).$$

Step 1: The plane. Let $d = 2$. Choose $R = Cr_n$ so large that $D_n \subset B(y_0, R)$; then $|y_0| + R \leq 2n$ for large n . Let $\chi_R(\zeta) := (1 + |\zeta|)^{-2} \mathbf{1}_{\{|\zeta| \leq R\}}$. For $x \in A_n$ and $v \in C_x$ we have $|y_0 - v| < R$ and, because $|y_0 - z| \leq \sqrt{d}/2$, also $|y_0 - v| \leq |x - z| + \sqrt{d}$. So comparing weights cell by cell gives $W_z \leq C(\chi_R * \tilde{\ell}_n)(y_0)$, and (5.8) gives

$$W_z \leq C(\chi_R * U_{D_n})(y_0) + C\delta \|\chi_R\|_1.$$

Write $U_{D_n} = U_{\{\Psi < b\}} + U_E$. By Lemma 2.2, $U_{\{\Psi < b\}} = 2d\varepsilon(b - \Psi)_+$ vanishes at y_0 and is $2d\varepsilon\Lambda_\Psi$ -Lipschitz, so $(\chi_R * U_{\{\Psi < b\}})(y_0) \leq 2d\varepsilon\Lambda_\Psi \int \chi_R(\zeta)|\zeta| d\zeta \leq Cr_n$. With (5.12) for $\chi = \chi_R$, the bounds $\|\chi_R\|_1 \leq 2\pi \log(1 + R) \leq C \log n$ and $H \leq \delta$ give, for large n ,

$$W_z \leq C(r_n + \delta \log n) \leq C(r_n + \sqrt{r_n}(\log n)^2) \leq Cr_n,$$

and (5.13) gives $H \leq C\sqrt{r_n \log n}$.

Step 2: A convolution kernel. Let $d \geq 3$. Convolutions are on $\mathbb{R}^d$, and

$$\chi(\zeta) := (1 + |\zeta|)^{2-2d},$$

For $d \geq 3$, this kernel is integrable, and splitting the convolution according to $|v| \geq |\zeta|/2$ or $|\zeta - v| \geq |\zeta|/2$ gives $\chi * \chi \leq A\chi$, with $A := 2^{2d-1}\|\chi\|_1$.

Step 3: Absorption. If y lies in the closure of C_x , then comparing weights cell by cell gives $W_x \leq C(\chi * \tilde{\ell}_n)(y)$. If moreover $y \in C_x$ and $|y| \leq 2n$, then (3.12), (3.11) and (3.13) give

$$\tilde{\ell}_n(y) \leq U_{D_n}(y) + C \log n + C\sqrt{(\chi * \tilde{\ell}_n)(y) \log n}.$$

For every fixed $\gamma > 0$, Young's inequality gives, on $B(0, 2n)$,

$$(5.14) \quad \tilde{\ell}_n \leq U_{D_n} + C_\gamma \log n + \gamma \chi * \tilde{\ell}_n.$$

Outside $B(0, 2n)$ we have $\tilde{\ell}_n = 0$ and $|U_{D_n}| \leq Cn^{2-d} \leq C$, because $|D_n| \leq n$ and $D_n \subset B(0, n + \sqrt{d})$; so (5.14) holds on $\mathbb{R}^d$ after increasing C_γ . With $\gamma := (2A)^{-1}$, convolving (5.14) with χ and using $\chi * \chi \leq A\chi$ and $\tilde{\ell}_n \geq 0$ gives $\chi * \tilde{\ell}_n \leq \chi * U_{D_n} + C \log n + 1/2 \chi * \tilde{\ell}_n$, where $\chi * \tilde{\ell}_n$ is bounded; hence

$$(5.15) \quad \chi * \tilde{\ell}_n \leq 2\chi * U_{D_n} + C \log n.$$

Step 4: The convolved potential at y_0 . We bound $\chi * U_{D_n}$ using $U_{D_n} = U_{\{\Psi < b\}} + U_E$. By Lemma 2.2, $U_{\{\Psi < b\}} = 2d\varepsilon(b - \Psi)_+$ is $2d\varepsilon\Lambda_\Psi$ -Lipschitz with values in $[0, 2d\varepsilon b]$, so $U_{\{\Psi < b\}}(y - \zeta) \leq U_{\{\Psi < b\}}(y) + 2d\varepsilon \min\{\Lambda_\Psi|\zeta|, b\}$. In polar coordinates, the integral of $\chi(\zeta) \min\{\Lambda_\Psi|\zeta|, b\}$ over $\mathbb{R}^d$ is at most $C \log(2 + b)$, which is at most $C \log n$ because $b \leq Cr_n$. Hence, for every $y \in \mathbb{R}^d$,

$$(5.16) \quad (\chi * U_{\{\Psi < b\}})(y) \leq \|\chi\|_1 U_{\{\Psi < b\}}(y) + C \log n.$$

Since $U_{\{\Psi < b\}}(y_0) = 0$, the bounds (5.15), (5.16) and (5.12) give $(\chi * \tilde{\ell}_n)(y_0) \leq C(H + \log n)$, and since y_0 lies in the closure of C_z , the first sentence of Step 3, with $x = z$ and $y = y_0$, gives

$$(5.17) \quad W_z \leq C(H + \log n).$$

With (5.13), the bound (5.17) gives $H \leq C\sqrt{(H + \log n) \log n} + C \log n \leq H/2 + C \log n$, so $H \leq C \log n$. $\square$

Lemma 5.2 and the consequence $\mathcal{I} \leq Cr_n^d H$ of (4.10) bound $\mathcal{I}$, hence $|E|$, and then (5.5) shows that b is close to r_n .

Proof of the bounds on the inner radius and the local times. We work on the event of Section 5.1 and take n large.

Step 1: The volume bound. Let $s := r_n q_n^{1/2}$. Split E into the shell $\{b \leq \Psi < b + s\}$, of volume at most $Cr_n^{d-1}s$, and its complement, of volume at most $\mathcal{I}/s$. By (4.10) and Lemma 5.2,

$$(5.18) \quad |E| \leq Cr_n^{d-1}s + \frac{\mathcal{I}}{s} \leq Cr_n^{d-1}s + \frac{Cr_n^{d+1}q_n}{s} = Cr_n^d q_n^{1/2}.$$

Step 2: The inner radius. By (5.5), (5.7) and $|X_n| \leq R_{\text{out}}(n) \leq Cr_n$,

$$\left| n - 2\varepsilon \sum_{x \in A_n} \Psi(x) \right| \leq Cr_n^{(d+3)/2} (\log n)^{1/2}.$$

Replacing the sum by $\int_{D_n} \Psi(v) dv$ changes the sum by at most $\Lambda_\Psi(\sqrt{d}/2)|A_n| \leq Cr_n^d$. Also

$$\int_{D_n} \Psi(v) dv = \frac{d|B_\Psi|}{d+1} b^{d+1} + \int_E \Psi(v) dv \quad \text{and} \quad 0 \leq \int_E \Psi(v) dv \leq Cr_n |E|.$$

Since $n = 2\varepsilon d|B_\Psi|r_n^{d+1}/(d+1)$ by (2.3), dividing by n gives

$$(5.19) \quad \left| 1 - \left(\frac{b}{r_n} \right)^{d+1} \right| \leq C(r_n^{(1-d)/2} (\log n)^{1/2} + r_n^{-1} + |E|r_n^{-d}).$$

By (5.18), and because $r_n^{(1-d)/2} (\log n)^{1/2}$ and r_n^{-1} are at most $q_n^{1/2}$, the right side is at most $Cq_n^{1/2}$, so

$$(5.20) \quad |b - r_n| \leq Cr_n q_n^{1/2}.$$

Discarding $-|X_n|^2$ in (5.5) and $\int_E \Psi(v) dv \geq 0$ before taking absolute values gives, without (5.18), the one-sided bound $(b - r_n)_+ \leq C(r_n^{(3-d)/2} (\log n)^{1/2} + 1)$.

Step 3: The local times. Lemma 2.2 gives $U_{D_n}(y) = 2d\varepsilon(b - \Psi(y))_+ + U_E(y)$ for every $y \in \mathbb{R}^d$, and (4.11), the consequence $\mathcal{I} \leq Cr_n^d H$ of (4.10) and Lemma 5.2 give $\sup |U_E| \leq C(r_n^{d+1} q_n)^{1/(d+1)} = Cr_n q_n^{1/(d+1)}$. With (5.8) and (5.20), these give, for $|y| \leq 2n$,

$$(5.21) \quad |\tilde{\ell}_n(y) - 2d\varepsilon(r_n - \Psi(y))_+| \leq \delta + Cr_n q_n^{1/(d+1)} + 2d\varepsilon|b - r_n| \leq Cr_n q_n^{1/(d+1)},$$

because $\delta = Cr_n^{1/2} \log n$ and $r_n q_n^{1/2}$ are at most $r_n q_n^{1/(d+1)}$; for $|y| > 2n$ both $\tilde{\ell}_n(y)$ and $2d\varepsilon(r_n - \Psi(y))_+$ vanish. $\square$

5.3. The outer radius. We bound $\max_{j \leq n} \Psi(X_j)$ by Lemma 4.3 in the direction of a Euclidean projection onto $\{\Psi \leq r_n\}$, with Π_n as in Section 5.1. Recall that, for $x \in \mathbb{R}^d$, every y with $\Psi(y) \leq r_n$ satisfies $(x - \Pi_n(x)) \cdot (y - \Pi_n(x)) \leq 0$.

Proof of Proposition 5.1. We work on the event of Section 5.1 and take n large. Then (5.20), in which $b = \inf_{y \notin D_n} \Psi(y)$, gives (5.1), and, since $\tilde{\ell}_n = \ell_n$ at lattice sites, (5.21) gives (5.3).

For the outer radius, let $w := \max_{0 \leq j \leq n} |X_j - \Pi_n(X_j)|$. If $w = 0$, then $\Psi(X_j) \leq r_n$ for $0 \leq j \leq n$, and (5.2) holds. Otherwise, let x_* be one of the sites $X_0, \dots, X_n$ with $|x_* - \Pi_n(x_*)| = w$, and let

$$u := \frac{x_* - \Pi_n(x_*)}{w}, \quad q := \Lambda_\Psi u \quad \text{and} \quad \beta := q \cdot \Pi_n(x_*).$$

The projection inequality gives $q \cdot y \leq \beta$ on $\{\Psi \leq r_n\}$, with strict inequality in its interior. In particular, $\beta \geq 0$, and

$$(5.22) \quad \Psi(x) \geq r_n \quad \text{for every } x \in \mathbb{R}^d \text{ with } q \cdot x \geq \beta.$$

Let $T := q \cdot x_* = \beta + \Lambda_\Psi w$. Since $|X_j - \Pi_n(X_j)| \leq w$ and $|q| = \Lambda_\Psi$, we have $q \cdot X_j \leq \beta + \Lambda_\Psi w = T$ for $0 \leq j \leq n$. Let $\theta := c_\Psi^2 / (8\Lambda_\Psi^2)$, and let y be one of the sites $X_0, \dots, X_n$ with $q \cdot y > T - \theta\Lambda_\Psi w$. Then $q \cdot y > \beta$, so $\Psi(y) > r_n$, because $\Psi(y) \leq r_n$ would give $q \cdot y \leq \beta$; hence $w_y := |y - \Pi_n(y)|$ is positive; let $v_y := (y - \Pi_n(y))/w_y$. Since $q \cdot \Pi_n(y) \leq \beta$, we get $\Lambda_\Psi w_y u \cdot v_y \geq q \cdot y - \beta > (1 - \theta)\Lambda_\Psi w$, and since $w_y \leq w$, this gives $u \cdot v_y > 1 - \theta$, so $|u - v_y| < (2\theta)^{1/2}$. Let $\xi \in \partial\Psi(y)$. The subgradient inequality at y and $\Psi(\Pi_n(y)) \leq r_n$ give $\xi \cdot (y - \Pi_n(y)) \geq \Psi(y) - r_n$; since $\Psi(r_n y / \Psi(y)) = r_n$ and w_y is the distance from y to $\{\Psi \leq r_n\}$, also $w_y \leq |y|(1 - r_n/\Psi(y)) \leq (\Psi(y) - r_n)/c_\Psi$. Hence $\xi \cdot v_y \geq c_\Psi$, and, because $|\xi| \leq \Lambda_\Psi$,

$$q \cdot \xi \geq \Lambda_\Psi (\xi \cdot v_y - \Lambda_\Psi |u - v_y|) \geq \Lambda_\Psi (c_\Psi - \Lambda_\Psi (2\theta)^{1/2}) = \frac{\Lambda_\Psi c_\Psi}{2}.$$

By (5.22) and (5.21), every $x \in \mathbb{Z}^d$ with $q \cdot x \geq \beta$ satisfies $\ell_n(x) \leq Cr_n q_n^{1/(d+1)}$. Since $x_* \in \mathbb{Z}^d$, $|x_*| \leq n$ and $\Psi(x_*) > r_n$, the bound (4.3) holds for q and every $a \in \Lambda_\Psi \mathbb{Z} \cap [0, \Lambda_\Psi n]$; so Lemma 4.3 with $h = \theta\Lambda_\Psi w$, $\alpha = \Lambda_\Psi c_\Psi/2$ and β in place of b gives

$$T \leq \max\{T - \theta\Lambda_\Psi w, \beta\} + Cr_n q_n^{1/(d+1)} \log n.$$

Since $T - \theta\Lambda_\Psi w = \beta + (1 - \theta)\Lambda_\Psi w > \beta$, this gives $\theta\Lambda_\Psi w \leq Cr_n q_n^{1/(d+1)} \log n$. Finally, $\Psi(X_j) \leq \Psi(\Pi_n(X_j)) + \Lambda_\Psi |X_j - \Pi_n(X_j)| \leq r_n + \Lambda_\Psi w$ for $0 \leq j \leq n$, which proves (5.2). $\square$

Proof of Theorem 2.1. Apply Proposition 5.1 with $p > 1$ and the Borel–Cantelli lemma. Its errors are $o(r_n)$, so the inner- and outer-radius bounds give (i), and the local-time bound gives (ii). At every fixed site x , part (ii) gives $\ell_n(x)/r_n \rightarrow 2d\varepsilon > 0$, proving (iii). Taking the Euclidean norm gives Theorem 1.1. $\square$

The proofs in Sections 2–5 use the symmetry $\Psi(-x) = \Psi(x)$ only to bound the coordinates of the subgradients. Let $K \subset \mathbb{R}^d$ be a compact convex set with the origin in its interior, and let $\Psi_K(x) := \inf\{t > 0 : x \in tK\}$ be its Minkowski functional, which is convex and positively homogeneous and satisfies $c|x| \leq \Psi_K(x) \leq C|x|$. Every $\xi \in \partial\Psi_K(x)$ satisfies $-\Psi_K(-e_i) \leq \xi_i \leq \Psi_K(e_i)$; so, if $\varepsilon \max\{\Psi_K(e_i), \Psi_K(-e_i)\} < 1/d$ for $1 \leq i \leq d$, then the probabilities (2.1) are positive, and the proofs of Theorem 2.1 and Proposition 5.1 apply without change to Ψ_K , with $|K|$ in place of $|B_\Psi|$. Thus every such set K is the limit shape of the centrally excited random walk with Ψ_K in place of Ψ .

6. THE INNER RADIUS FOR THE EUCLIDEAN NORM

We prove the bounds on $|R_{\text{in}}(n) - r_n|$ and on the volume in Theorem 1.2 (i) and (ii), and a first uniform bound on the local times (Proposition 6.1). We use the inequality (6.3) below to turn the bound of Lemma 5.2 on the potential at the contact point y_0 , a point of the sphere $\{|v| = R_{\text{in}}(n)\}$ in the closure of a cell disjoint from D_n , into a bound on the volume of D_n outside the ball $B(0, R_{\text{in}}(n))$: every point v of D_n outside this ball contributes at least $c|v - y_0|^{2-d}/r_n$ to the potential at y_0 (Figure 5), a bound that does not vanish near the sphere $\{|v| = R_{\text{in}}(n)\}$.

Proposition 6.1 (Inner radius). *Let Ψ be the Euclidean norm, and let $0 < \varepsilon < 1/d$. For every $p > 0$ there exists $C(d, \varepsilon, p) < \infty$ such that, for every integer $n \geq 2$, with probability at least $1 - Cn^{-p}$, the following hold simultaneously, where q_n is defined in (5.4).*

(i) Inner radius:

$$(6.1) \quad |R_{\text{in}}(n) - r_n| \leq Cr_n q_n.$$

(ii) Volume:

$$|r_n^{-1} D_n \triangle B(0, 1)| \leq Cq_n.$$

(iii) Local times:

$$\sup_{y \in \mathbb{R}^d} |\tilde{\ell}_n(y) - 2d\varepsilon(r_n - |y|)_+| \leq Cr_n q_n^{1/d}.$$

Fix $p > 0$. In this section and the next we take n large and work on the event of Section 5.1; its complement has probability at most Cn^{-p} . With K as in Section 5.1, we have $D_n \subset B(0, Kr_n)$, and on this ball (3.3) and $\max_x \ell_n(x) \leq Cr_n$ give, with δ redefined for $d \geq 3$,

$$(6.2) \quad |\tilde{\ell}_n(y) - U_{D_n}(y)| \leq \delta, \quad \text{where} \quad \delta := \begin{cases} C\sqrt{r_n} \log n, & d = 2, \\ C\sqrt{r_n} \log n, & d \geq 3 \end{cases}.$$

Let $b := R_{\text{in}}(n)$ and $E := D_n \setminus B(0, b)$, as in Section 4.2 for the Euclidean norm, and let y_0, z and H be as there, so that

$$|y_0| = b, \quad \ell_n(z) = 0, \quad |z - y_0| \leq \sqrt{d}/2, \quad |z| \geq b, \quad H = U_{D_n}(y_0) = U_E(y_0).$$

6.1. The volume outside the inner ball. If $s = |v| \geq b = |y_0|$, then

$$(6.3) \quad \frac{u_v \cdot (v - y_0)}{|v - y_0|^d} = \frac{s^2 - v \cdot y_0}{s|v - y_0|^d} \geq \frac{1}{2s}|v - y_0|^{2-d} \geq 2^{1-d}s^{1-d}.$$

The first inequality uses $2(s^2 - v \cdot y_0) = |v - y_0|^2 + s^2 - b^2$, and the second $|v - y_0| \leq 2s$ and $d \geq 2$. Since $|v| \leq Cr_n$ on E , integrating (6.3) over E gives $H \geq c|E|r_n^{1-d}$, and Lemma 5.2 gives

$$(6.4) \quad |E| \leq Cr_n^{d-1}H \leq Cr_n^d q_n.$$

6.2. Upper bounds on the local times. For a bounded measurable set $D \subset \mathbb{R}^d$ and $y \in \mathbb{R}^d$, the potential of D with its negative contributions discarded is

$$U_D^+(y) := \frac{2\varepsilon}{\omega_d} \int_D \left[u_v \cdot \frac{v - y}{|v - y|^d} \right]_+ dv.$$

Suppose that $d \geq 3$, let χ and A be as in Step 2 of the proof of Lemma 5.2, and let m be the function in (5.10) for this χ . Since $0 \leq m \leq \|\chi\|_1$, discarding the points v with $u_v \cdot (v - y) < 0$ in (5.11) with $D = E$ gives $\chi * U_E \leq \|\chi\|_1 U_E^+$; also $|\chi * U_E| \leq \|\chi\|_1 \sup |U_E|$. With (5.16), these two bounds give, for every $y \in \mathbb{R}^d$,

$$(6.5) \quad (\chi * U_{D_n})(y) \leq \|\chi\|_1 (U_{B(0,b)}(y) + \min\{U_E^+(y), \sup |U_E|\}) + C \log n.$$

Since $U_E \leq \min\{U_E^+, \sup |U_E|\}$, the bounds (5.14) with $\gamma = (2A)^{-1}$, (5.15) and (6.5) give the first of the bounds

$$(6.6) \quad \tilde{\ell}_n(y) \leq C((b - |y|)_+ + \sup |U_E| + \log n) \quad \text{and} \quad \tilde{\ell}_n(y) \leq C(U_E^+(y) + \log n)$$

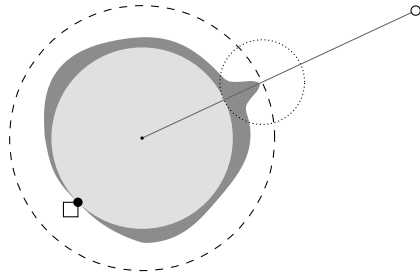


FIGURE 5. The light disk is $B(0, b)$, the largest ball about the origin in D_n , the dark region is $E = D_n \setminus B(0, b)$, and the dashed circle, of radius $b + w$, contains D_n . The square is a cell disjoint from D_n whose closure contains the point y_0 (black dot) of the circle $\{|v| = b\}$, and every point of E contributes nonnegatively to $U_E(y_0)$. The segment joins the origin (small dot) to $y_t = (b + w + 4t)u$ (open dot) for a unit vector u , and the points of E in the dotted ball $B((b + w)u, t)$ contribute negatively to $U_E(y_t)$.

for every $y \in \mathbb{R}^d$, and the second for $|y| \geq b$, where $U_{B(0,b)}(y) = 0$.

Suppose that $d = 2$. By (6.2), (1.10), (3.14) and (6.4), every $y \in \mathbb{R}^2$ satisfies

$$(6.7) \quad \tilde{\ell}_n(y) \leq 2d\varepsilon(b - |y|)_+ + Cr_n^{3/4}(\log n)^{1/4},$$

since $\sqrt{r_n} \log n = o(r_n^{3/4}(\log n)^{1/4})$ and $\tilde{\ell}_n$ vanishes outside $B(0, Kr_n)$.

Let $L := 1 + \max\{\ell_n(x) : x \in \mathbb{Z}^d, |x| \geq b\}$. By (6.6), (3.14) and (6.4) when $d \geq 3$, and by (6.7) when $d = 2$,

$$(6.8) \quad L \leq Cr_n q_n^{1/d}.$$

6.3. The inner radius.

Proof of Proposition 6.1. By (5.19) and (6.4),

$$\left| 1 - \left(\frac{b}{r_n} \right)^{d+1} \right| \leq C(r_n^{(1-d)/2}(\log n)^{1/2} + r_n^{-1} + q_n) \leq Cq_n,$$

because $r_n^{(1-d)/2}(\log n)^{1/2} \leq q_n$. This proves (6.1). Since D_n is the disjoint union of $B(0, b)$ and E , the bounds (6.1) and (6.4) prove (ii).

On $B(0, Kr_n)$, the bounds (6.2), (1.10), (3.14) and (6.1) give

$$|\tilde{\ell}_n(y) - 2d\varepsilon(r_n - |y|)_+| \leq C(\delta + r_n q_n + |E|^{1/d}) \leq Cr_n q_n^{1/d},$$

and outside $B(0, Kr_n)$ both $\tilde{\ell}_n(y)$ and $2d\varepsilon(r_n - |y|)_+$ vanish, which proves (iii). $\square$

7. THE OUTER RADIUS AND THE LOCAL TIMES

We prove the bound on $R_{\text{out}}(n)$ in Theorem 1.2(i) (Proposition 7.1) and the bound on the local times in Theorem 1.2(iii). We bound the largest local time outside the ball $B(0, R_{\text{in}}(n))$; the crossing estimate (4.7) then bounds $R_{\text{out}}(n) - R_{\text{in}}(n)$ by a constant times $\log n$ times one plus this local time. To bound this local time, we bound the volume of $E = D_n \setminus B(0, R_{\text{in}}(n))$ near the sphere of radius $R_{\text{out}}(n) + 2\sqrt{d}$ about the origin. For a unit vector u and a number t with $R_{\text{out}}(n) - R_{\text{in}}(n) + 2\sqrt{d} \leq t \leq r_n/4$, Step 1 of the proof of (7.5) evaluates the potential at the point $y_t = (R_{\text{out}}(n) + 2\sqrt{d} + 4t)u$ outside this sphere. The cell containing y_t is disjoint from D_n , and its center is a site outside A_n . Dynkin's formula at this center, with a bound on the bracket of the martingale, bounds U_{D_n} at this center, and (3.13) transfers this bound to y_t ; the resulting bound on $|U_E(y_t)|$ is (7.6). The points of E in the ball of radius t about $(R_{\text{out}}(n) + 2\sqrt{d})u$, the point of the sphere closest to y_t , contribute negatively to $U_E(y_t)$. Since $|U_E(y_t)|$ and the positive contributions to $U_E(y_t)$ are small, the volume of E in this ball is small; the resulting bound on this volume is (7.7). Step 2 of the proof of (7.11) uses the same point y_t when $d = 2$.

Proposition 7.1 (Outer radius). *Let Ψ be the Euclidean norm, and let $0 < \varepsilon < 1/d$. For every $p > 0$ there exists $C(d, \varepsilon, p) < \infty$ such that, for every integer $n \geq 2$, with probability at least $1 - Cn^{-p}$,*

$$R_{\text{out}}(n) \leq r_n + C \begin{cases} \sqrt{r_n}(\log n)^{5/2}, & d = 2, \\ (\log n)^{d+1}, & d \geq 3 \end{cases}.$$

With $b = R_{\text{in}}(n)$, $E = D_n \setminus B(0, b)$ and L as in Section 6, let

$$w := R_{\text{out}}(n) - b + 2\sqrt{d},$$

so that $w \geq 3\sqrt{d}/2 \geq 1$, because $b \leq R_{\text{out}}(n) + \sqrt{d}/2$, and $E \subset \{b \leq |v| \leq b + w\}$. Then $b \asymp r_n$ and $b + w \asymp r_n$, by (6.1) and Proposition 4.1.

On the event of Section 5.1, the bound (4.7) with $a = b = R_{\text{in}}(n)$ gives $R_{\text{out}}(n) \leq b + CL \log n$. Because $L \geq 1$, this gives

$$(7.1) \quad w \leq CL \log n,$$

and with (6.8) the first bound

$$(7.2) \quad w \leq Cr_n q_n^{1/d} \log n = o(r_n (\log n)^{-d}).$$

7.1. The potential beyond the inner radius. Since E lies in the annulus $\{b \leq |v| \leq b + w\}$, Lemma 3.4 with $a = w$ gives

$$(7.3) \quad \sup_{y \in \mathbb{R}^d} |U_E(y)| \leq 2d\varepsilon w.$$

Both dimensions use the following deterministic bound on the positive contributions to the potential at a point between the two radii, in terms of the volume near the outer sphere.

Lemma 7.2. *There exists $C(d, \varepsilon) < \infty$ with the following property. Let $1 \leq w \leq b$ and $\lambda \geq 1$, and let D be a measurable subset of $\{v \in \mathbb{R}^d : b \leq |v| \leq b + w\}$ such that $|D \cap B((b + w)u, t)| \leq \lambda t^{d-1}$ for every unit vector u and every $t \geq w$, and $\int_D |v - y|^{2-d} dv \leq \lambda b$ for every $y \in \mathbb{R}^d$. Then every $y \in \mathbb{R}^d$ with $b \leq |y| \leq b + w$ satisfies*

$$U_D^+(y) \leq C(\lambda w^{d-1})^{1/d} + C\lambda.$$

Proof. Fix y with $b \leq |y| \leq b + w$. If $t \geq w$, then $B(y, t) \subset B((b + w)u_y, 2t)$, so $|D \cap B(y, t)| \leq 2^{d-1} \lambda t^{d-1}$. In particular, the points of D within distance w of y have total volume at most $2^{d-1} \lambda w^{d-1}$, so, as in the proof of (3.14), the integral of $|v - y|^{1-d}$ over them is at most $C(\lambda w^{d-1})^{1/d}$. For the other points $v \in D$, both $|v|$ and $|y|$ lie in $[b, b + w]$, so the identity $2(|v|^2 - v \cdot y) = |v - y|^2 + |v|^2 - |y|^2$ and $|v|^2 - |y|^2 \leq 3bw$ give

$$\left[u_v \cdot \frac{v - y}{|v - y|^d} \right]_+ \leq \frac{2w}{|v - y|^d} + \frac{1}{2b|v - y|^{d-2}}.$$

The second term contributes at most $C\lambda$ to $U_D^+(y)$, by the second hypothesis, and integrating the first by parts against the bound on $|D \cap B(y, t)|$ shows that it contributes at most $Cw \int_w^\infty \lambda t^{d-1} t^{-d-1} dt \leq C\lambda$. $\square$

7.2. The outer radius in higher dimensions. Suppose that $d \geq 3$. Because the sphere of radius $b + w$ is curved, the positive contributions of E to the potential at a point outside this sphere are at most a constant times

$$(7.4) \quad \Theta := \log n + \frac{1}{r_n} \sup_{y \in \mathbb{R}^d} \int_E |v - y|^{2-d} dv.$$

We prove that

$$(7.5) \quad w \leq C(\log n)^{d+1}.$$

By (7.1) and (6.6), it suffices to bound $U_E^+(y)$ for $b \leq |y| \leq b + w$.

Proof of (7.5). Step 1: The potential at a point of a cell disjoint from D_n . If $|y| \geq b + w$, then every $v \in E$ satisfies $|v| \leq |y|$, so the identity $2(|v|^2 - v \cdot y) = |v - y|^2 + |v|^2 - |y|^2$ gives $u_v \cdot (v - y) \leq |v - y|^2 / (2|v|)$; since $|v| \geq b \geq cr_n$ on E , the definition (7.4) gives $U_E^+(y) \leq C\Theta$. Let u be a unit vector, let $w \leq t \leq r_n/4$, let $y_t := (b + w + 4t)u$ (Figure 5), and let $z \in \mathbb{Z}^d$ be the site with $y_t \in C_z$. Then $|z| \geq |y_t| - \sqrt{d}/2 \geq b + w > R_{\text{out}}(n)$, so $\ell_n(z) = 0$, and the first sentence of Step 3 of the proof of Lemma 5.2, with $x = y = z$, together with (5.15), (6.5) and $U_{B(0,b)}(z) = 0$, gives

$W_z \leq C(\chi * \tilde{\ell}_n)(z) \leq C\Theta$, where W_z is the sum in (3.7). Hence (3.12) and (3.11), which apply at z because $|z| \leq Cr_n < 3n$, together with (3.13), $U_{B(0,b)}(y_t) = 0$ and $\Theta \geq \log n$, give

$$(7.6) \quad |U_E(y_t)| = |U_{D_n}(y_t)| \leq C(\sqrt{\Theta \log n} + \log n) \leq C\Theta.$$

Step 2: The volume of E near the sphere of radius $b + w$. We show that, for every unit vector u and every $t \geq w$,

$$(7.7) \quad |E \cap B((b+w)u, t)| \leq C\Theta t^{d-1}.$$

For $t > r_n/4$ this follows from $|E| \leq Cr_n^{d-1} \log n$, which is (6.4) for $d \geq 3$, and $\Theta \geq \log n$, so let $w \leq t \leq r_n/4$, with y_t as in Step 1. By Step 1, the positive parts of $u_v \cdot (v - y_t)|v - y_t|^{-d}$ contribute at most $C\Theta$ to $U_E(y_t)$. If $v \in E \cap B((b+w)u, t)$, then $u_v \cdot u \geq (b+w-t)/(b+w)$, hence

$$u_v \cdot (v - y_t) = |v| - (b+w+4t)u_v \cdot u \leq (b+w) - (b+w+4t) \left(1 - \frac{t}{b+w}\right) \leq -t,$$

where the last inequality uses $t \leq r_n/4 \leq (b+w)/2$, which holds for large n by (6.1). Since also $|v - y_t| \leq 5t$, we get $u_v \cdot (v - y_t)|v - y_t|^{-d} \leq -ct^{1-d}$ for each such v . Thus $U_E(y_t) \leq C\Theta - ct^{1-d}|E \cap B((b+w)u, t)|$, which with (7.6) proves (7.7).

Step 3: The largest local time beyond the inner radius. If $t \geq w$ and $b \leq |y| \leq b+w$, then $B(y, t) \subset B((b+w)u_y, 2t)$, so (7.7) gives

$$(7.8) \quad |E \cap B(y, t)| \leq C\Theta t^{d-1}.$$

By (7.7) and the definition (7.4) of Θ , Lemma 7.2 applies to $D = E$ with $\lambda = C\Theta$ (here $w \leq b$ by (7.2)), and gives $U_E^+(y) \leq C(\Theta w^{d-1})^{1/d} + C\Theta$ for $b \leq |y| \leq b+w$. Every site $x \in A_n$ with $|x| \geq b$ satisfies $|x| \leq b+w$, so (6.6) and $\Theta \geq \log n$ give

$$(7.9) \quad L \leq C((\Theta w^{d-1})^{1/d} + \Theta).$$

Step 4: A bound on Θ . Let $y \in \mathbb{R}^d$. If $t \geq w$ and $B(y, t)$ meets E at a point v_1 , then $B(y, t) \subset B(v_1, 2t)$ and $|v_1| \in [b, b+w]$; so (7.8) and the bound $|E| \leq Cr_n^{d-1} \log n$ of (6.4) give $|E \cap B(y, t)| \leq C \min\{\Theta t^{d-1}, r_n^{d-1} \log n\}$ for $t \geq w$, while $|E \cap B(y, t)| \leq \omega_d t^d$ for $t < w$. Fubini's theorem and a split at $t = r_n(\log n / \Theta)^{1/(d-1)}$, where the two terms of the minimum agree, give

$$\int_E |v - y|^{2-d} dv = (d-2) \int_0^\infty |E \cap B(y, t)| t^{1-d} dt \leq Cw^2 + Cr_n \Theta^{(d-2)/(d-1)} (\log n)^{1/(d-1)}.$$

By (7.4) and Young's inequality, $\Theta \leq \Theta/2 + C(w^2/r_n + \log n)$, so

$$(7.10) \quad \Theta \leq C(w^2/r_n + \log n).$$

Step 5: The bound on w . By (7.1), (7.9) and Young's inequality,

$$w \leq C \log n ((\Theta w^{d-1})^{1/d} + \Theta) \leq \frac{w}{2} + C\Theta (\log n)^d,$$

so $w \leq C\Theta (\log n)^d$, and (7.10) gives

$$w \leq C(\log n)^d w^2/r_n + C(\log n)^{d+1}.$$

By (7.2), $C(\log n)^d w/r_n \leq 1/2$ for large n , so the first term on the right is at most $w/2$, which proves (7.5). $\square$

7.3. The outer radius in the plane. Suppose that $d = 2$. We prove that

$$(7.11) \quad w \leq C\sqrt{r_n}(\log n)^{5/2}.$$

The argument follows Section 7.2 with $\sqrt{r_n \log n}$ in place of Θ , except that Step 1 bounds the sum W_x of (3.7) directly, since (6.6) requires $d \geq 3$.

Proof of (7.11). Step 1: Local times beyond the inner radius. By (6.2), (1.10) and (7.3), every $x \in \mathbb{Z}^2$ satisfies

$$\ell_n(x) \leq 2d\varepsilon(b - |x|)_+ + Cw + C\sqrt{r_n} \log n.$$

Let $x \in \mathbb{Z}^2$ satisfy $b \leq |x| \leq 3n$. Since the local times vanish off A_n , the sum W_x runs over $y \in A_n \subset B(0, Kr_n)$; bound each local time $\ell_n(y)$ in it by this inequality: the terms $2d\varepsilon(b - |y|)_+$ contribute at most Cr_n to W_x , because $(b - |y|)_+ \leq |y - x|$, and the other two terms at most $C(w + \sqrt{r_n} \log n) \log n = o(r_n)$, by (7.2). Hence $W_x \leq Cr_n$, so (3.11) gives $|\mathcal{M}_n^x| \leq C\sqrt{r_n \log n}$, and (3.12) and $U_{B(0,b)}(x) = 0$ give

$$(7.12) \quad |\ell_n(x) - U_E(x)| \leq C\sqrt{r_n \log n}.$$

Step 2: Volume bounds and a bound on U_E^+ . For $y_t = (b + w + 4t)u$, where u is a unit vector and $w \leq t \leq r_n/4$, the inequality $u_v \cdot (v - y_t) \leq |v - y_t|^2/(2|v|)$ of Step 1 of Section 7.2 bounds the positive contributions to $U_E(y_t)$ by $C \int_E (2|v|)^{-1} dv \leq C|E|/r_n \leq C\sqrt{r_n \log n}$, by (6.4). At the site z with $y_t \in C_z$, the bound (7.12), with $\ell_n(z) = 0$ and (3.13), gives $|U_E(y_t)| \leq C\sqrt{r_n \log n}$. For each $v \in E \cap B((b + w)u, t)$, as in Step 2 of Section 7.2, $u_v \cdot (v - y_t)|v - y_t|^{-2} \leq -ct^{-1}$, so, for every $t \geq w$,

$$|E \cap B((b + w)u, t)| \leq Ct\sqrt{r_n \log n},$$

where the case $t > r_n/4$ follows from $|E| \leq Cr_n^{3/2}(\log n)^{1/2}$. Lemma 7.2 applies to $D = E$ with $\lambda = C\sqrt{r_n \log n}$, because $|E| \leq Cr_n^{3/2}(\log n)^{1/2} \leq \lambda b$, and gives, for every y with $b \leq |y| \leq b + w$,

$$(7.13) \quad U_E^+(y) \leq C(r_n \log n)^{1/4}\sqrt{w} + C\sqrt{r_n \log n}.$$

Step 3: The bound on w . Since $U_E \leq U_E^+$, the bound (7.12) bounds L by a constant times the right side of (7.13), and (7.1) gives

$$w \leq C \log n \left((r_n \log n)^{1/4}\sqrt{w} + \sqrt{r_n \log n} \right) \leq \frac{w}{2} + C\sqrt{r_n}(\log n)^{5/2},$$

which proves (7.11). $\square$

Proof of Proposition 7.1. On the event of Section 5.1, for large n , we have $R_{\text{out}}(n) \leq b + w$ and, by (6.1), $b - r_n \leq Cr_n q_n$, which is $C \log n$ when $d \geq 3$ and $C\sqrt{r_n \log n}$ when $d = 2$. So (7.5) and (7.11) prove the proposition. $\square$

7.4. The uniform bound on the local times. We prove Theorem 1.2(iii) on the event of Section 5.1. Suppose first that $d \geq 3$. By (7.3) and (7.5), $\sup |U_E| \leq Cw \leq C(\log n)^{d+1}$, which with (6.2), (1.10) and (6.1) gives

$$\max_{y \in \mathbb{Z}^d, |y| \leq Kr_n} |\ell_n(y) - 2d\varepsilon(r_n - |y|)_+| \leq C(\sqrt{r_n \log n} + (\log n)^{d+1} + |b - r_n|) \leq C\sqrt{r_n \log n}$$

for large n , since $|b - r_n| \leq C \log n$; outside $B(0, Kr_n)$ both $\ell_n(y)$ and $2d\varepsilon(r_n - |y|)_+$ vanish.

Let $d = 2$. The function U_E is continuous (Section 3.2), harmonic in $B(0, b)$ because E does not meet that ball and each coordinate of $(v - y)|v - y|^{-2} = -\nabla_y \log |v - y|$ is harmonic in $y \neq v$, and nonnegative on the closure of $B(0, b)$ because $u_v \cdot (v - y) \geq 0$ when $|v| \geq b \geq |y|$. The maximum principle, the inequality $U_E \leq U_E^+$, (7.13) and (7.11) give

$$\sup_{|y| \leq b+w} U_E(y) \leq C\sqrt{r_n}(\log n)^{3/2}.$$

For $|y| \leq b$, the bounds (6.2) and (1.10) give

$$|\tilde{\ell}_n(y) - 2d\varepsilon(r_n - |y|)_+| \leq C(\sqrt{r_n}(\log n)^{3/2} + |b - r_n|).$$

For $|y| \geq b$, the absolute value of this difference is at most the larger of $\tilde{\ell}_n(y)$ and $2d\varepsilon(r_n - |y|)_+$, which are nonnegative. Here $2d\varepsilon(r_n - |y|)_+ \leq 2d\varepsilon|b - r_n|$, and $\tilde{\ell}_n(y)$ is at most $U_E(y) + C\sqrt{r_n} \log n$ by (6.2) if $|y| \leq b + w$ and vanishes otherwise. Since $|b - r_n| \leq C\sqrt{r_n} \log n$, restricting to sites proves Theorem 1.2 (iii) for $d = 2$.

Proof of Theorem 1.2. On the event of Section 5.1, for large n , Proposition 6.1 gives the bounds on $R_{\text{in}}(n)$ and on the volume, Proposition 7.1 the bound on $R_{\text{out}}(n)$, and Section 7.4 the bound on the local times. The Borel–Cantelli lemma with $p > 1$ gives the almost-sure statement. $\square$

8. LIMIT LAWS

We prove central limit theorems and laws of the iterated logarithm for $\sum_{x \in A_n} |x|$ and for the local times at fixed sites. After centering, these quantities are, up to smaller errors, multiples of the quadratic martingale $\mathcal{Q}$ of (5.5) and of the martingales $\mathcal{M}^y$ of Section 3.1; we compute the limits of their brackets from Theorem 1.1 (ii), and then apply the martingale central limit theorem and the law of the iterated logarithm.

8.1. The sum of the distances from the origin. For the Euclidean norm, (5.5) reads

$$(8.1) \quad \mathcal{Q}_n = 2\varepsilon \sum_{x \in A_n} |x| - n + |X_n|^2.$$

Let $R_{\text{mom}}(n)$ be the radius of the ball about the origin on which the integral of $|v|$ equals $\sum_{x \in A_n} |x|$, that is,

$$(8.2) \quad R_{\text{mom}}(n) := \left(\frac{d+1}{d\omega_d} \sum_{x \in A_n} |x| \right)^{1/(d+1)}.$$

Theorem 8.1 (Sum of the distances from the origin). *Let Ψ be the Euclidean norm, let $0 < \varepsilon < 1/d$, and let $R_{\text{mom}}(n)$ be as in (8.2). Then the following hold.*

(i) Central limit theorem: as $n \rightarrow \infty$,

$$\frac{\sum_{x \in A_n} |x| - n/(2\varepsilon)}{r_n^{(d+3)/2}} \xrightarrow{d} \mathcal{N}\left(0, \frac{2d\omega_d}{\varepsilon(d+2)(d+3)}\right)$$

and

$$\frac{R_{\text{mom}}(n) - r_n}{r_n^{(3-d)/2}} \xrightarrow{d} \mathcal{N}\left(0, \frac{2}{\varepsilon d\omega_d(d+2)(d+3)}\right).$$

(ii) Law of the iterated logarithm: for each choice of sign, almost surely,

$$(8.3) \quad \begin{aligned} \limsup_{n \rightarrow \infty} \frac{\pm(\sum_{x \in A_n} |x| - n/(2\varepsilon))}{r_n^{(d+3)/2} \sqrt{2 \log \log n}} &= \left(\frac{2d\omega_d}{\varepsilon(d+2)(d+3)} \right)^{1/2}, \\ \limsup_{n \rightarrow \infty} \frac{\pm(R_{\text{mom}}(n) - r_n)}{r_n^{(3-d)/2} \sqrt{2 \log \log n}} &= \left(\frac{2}{\varepsilon d\omega_d(d+2)(d+3)} \right)^{1/2}. \end{aligned}$$

In the plane, comparing $R_{\text{mom}}(n)$ with the inner and outer radii gives Theorem 1.3 (ii) and, with a lower bound on the probability that $\mathcal{Q}_n$ deviates by about $(\log n)^{1/2}$ standard deviations (Lemma 9.1), Theorem 1.3 (i).

Proof of Theorem 8.1. Step 1: The bracket. Under both (2.1) and the simple random walk transition probabilities, the conditional second-moment matrix of $X_{j+1} - X_j$ given $\mathcal{F}_j$ is $1/d$ times the identity, so for every $\mathcal{F}_j$ -measurable random vector q the conditional variance of $q \cdot (X_{j+1} - X_j)$ is $|q|^2/d - \varepsilon^2 I_j(q \cdot u_{X_j})^2$. Since $\mathcal{Q}_{j+1} - \mathcal{Q}_j = 2X_j \cdot (X_{j+1} - X_j + \varepsilon I_j u_{X_j})$, the bracket of $\mathcal{Q}$ is

$$(8.4) \quad \langle \mathcal{Q} \rangle_n = \frac{4}{d} \sum_{x \in \mathbb{Z}^d} \ell_n(x) |x|^2 - 4\varepsilon^2 \sum_{x \in A_n} |x|^2.$$

By Theorem 1.1 (i), almost surely, for all large n , $A_n \subset B(0, 2r_n)$, and hence $R_{\text{out}}(n) \leq 2r_n + 1$. With Theorem 1.1 (ii), this gives, almost surely,

$$\frac{1}{r_n^{d+3}} \sum_{x \in \mathbb{Z}^d} \ell_n(x) |x|^2 \longrightarrow 2d\varepsilon \int_{|v| < 1} (1 - |v|) |v|^2 dv = \frac{2d^2 \varepsilon \omega_d}{(d+2)(d+3)}.$$

The second sum in (8.4) is $O(r_n^{d+2})$, so

$$(8.5) \quad \frac{\langle \mathcal{Q} \rangle_n}{r_n^{d+3}} \longrightarrow \frac{8d\varepsilon \omega_d}{(d+2)(d+3)} \quad \text{almost surely.}$$

Step 2: The central limit theorem. The increment bound $|\mathcal{Q}_{j+1} - \mathcal{Q}_j| \leq C|X_j|$ and $R_{\text{out}}(n) = O(r_n)$ verify the conditional Lindeberg condition at scale $r_n^{(d+3)/2}$. By (8.5) and Hall and Heyde [1980, Corollary 3.1],

$$\frac{\mathcal{Q}_n}{r_n^{(d+3)/2}} \xrightarrow{d} \mathcal{N}\left(0, \frac{8d\varepsilon \omega_d}{(d+2)(d+3)}\right).$$

Step 3: The law of the iterated logarithm. The predictable increment bound $C|X_{n-1}|$ satisfies Stout's condition: its ratio to $\sqrt{\langle \mathcal{Q} \rangle_n / \log \log \langle \mathcal{Q} \rangle_n}$ is $O(\sqrt{\log \log n / n})$. Thus Stout [1970] applies to both $\mathcal{Q}$ and $-\mathcal{Q}$, and (8.5) gives the required normalization.

Step 4: The sum and the radius. In (8.1), $|X_n|^2 = O(r_n^2)$ is negligible on both scales, giving the two limit laws for the sum. They imply $R_{\text{mom}}(n)/r_n \rightarrow 1$ almost surely. Expanding (8.2) at $n/(2\varepsilon)$ gives

$$\frac{R_{\text{mom}}(n) - r_n}{r_n^{(3-d)/2}} = \left(\frac{1}{d\omega_d} + o(1) \right) \frac{\sum_{x \in A_n} |x| - n/(2\varepsilon)}{r_n^{(d+3)/2}}.$$

Slutsky's lemma and the positive limiting factor give the assertions for $R_{\text{mom}}(n)$. $\square$

8.2. The potential at a fixed site. Since $B(0, R_{\text{in}}(n)) \subset D_n \subset B(0, R_{\text{out}}(n) + \sqrt{d})$, Theorem 1.2 (i) and (ii) give, almost surely for all large n , $|D_n \triangle B(0, r_n)| \leq C r_n^d q_n$ and, for every $v \in D_n \triangle B(0, r_n)$,

$$(8.6) \quad ||v| - r_n| \leq C \begin{cases} \sqrt{r_n} (\log n)^{5/2}, & d = 2, \\ (\log n)^{d+1}, & d \geq 3 \end{cases}.$$

The product of q_n and the right side of (8.6) is $C(\log n)^3$ when $d = 2$ and $C(\log n)^{d+2}/r_n$ when $d \geq 3$. On the annulus defined by (8.6), the kernel $|v|^{1-d}$ of $U_{D_n}(0)$ is close to $r_n^{-d}|v|$, which relates $U_{D_n}(0)$ to $\sum_{x \in A_n} |x|$.

Lemma 8.2. *Let Ψ be the Euclidean norm, let $0 < \varepsilon < 1/d$, and let $y \in \mathbb{Z}^d$. There exists $C(d, \varepsilon, y) < \infty$ such that, almost surely, for all sufficiently large n ,*

$$(8.7) \quad \left| U_{D_n}(y) - 2d\varepsilon(r_n - |y|) - \frac{\mathcal{Q}_n}{\omega_d r_n^d} \right| \leq C \begin{cases} (\log n)^3, & d = 2, \\ 1, & d \geq 3 \end{cases},$$

where $\mathcal{Q}$ is the martingale of (5.5).

By Theorem 8.1, $\mathcal{Q}_n/(\omega_d r_n^d)$ has order $r_n^{(3-d)/2}$, smaller than the order of $\mathcal{M}_n^y$ by a factor $(\log r_n)^{1/2}$ in the plane and $r_n^{(d-2)/2}$ when $d \geq 3$.

Proof. Let $\nu := \mathbf{1}_{D_n} - \mathbf{1}_{B(0, r_n)}$ and $U_\nu(y) := (2\varepsilon/\omega_d) \int \nu(v) u_v \cdot (v - y) |v - y|^{-d} dv$, so that $U_{D_n} = U_{B(0, r_n)} + U_\nu$ and $\|\nu\|_1 \leq Cr_n^d q_n$. For large n the right side of (8.6) is at most $r_n/2$, so $|v| \geq r_n/2$ wherever $\nu(v) \neq 0$.

At the origin, the kernel is $u_v \cdot v |v|^{-d} = |v|^{1-d}$. The function $s \mapsto s^{1-d} - r_n^{-d} s$ vanishes at $s = r_n$ and has derivative $O(r_n^{-d})$ for $s \geq r_n/2$, so $||v|^{1-d} - r_n^{-d} |v|| \leq Cr_n^{-d} ||v| - r_n|$ for $|v| \geq r_n/2$. Integrating against ν , with (8.6) and the bound on $\|\nu\|_1$, and using $U_{B(0, r_n)}(0) = 2d\varepsilon r_n$ and $\int_{B(0, r_n)} |v| dv = n/(2\varepsilon)$, gives

$$\left| U_{D_n}(0) - 2d\varepsilon r_n - \frac{2\varepsilon}{\omega_d r_n^d} \left(\int_{D_n} |v| dv - \frac{n}{2\varepsilon} \right) \right| \leq C \begin{cases} (\log n)^3, & d = 2, \\ 1, & d \geq 3 \end{cases}.$$

Replacing the integral by $\sum_{x \in A_n} |x|$ changes the expression in parentheses by $O(r_n^d)$, and (8.1) proves (8.7) at $y = 0$, since $|X_n|^2/(\omega_d r_n^d) = O(r_n^{2-d}) = O(1)$.

For fixed y , the gradient of $u_v \cdot (v - y) |v - y|^{-d}$ with respect to y is $O_y(r_n^{-d})$ for $|v| \geq r_n/2$, so $|U_\nu(y) - U_\nu(0)| = O_y(r_n^{-d} \|\nu\|_1) = O_y(1)$. With $U_{B(0, r_n)}(y) - U_{B(0, r_n)}(0) = -2d\varepsilon|y|$, from (1.10), this proves (8.7). $\square$

8.3. Local times at fixed sites. Up to a smaller correction at first departures, the bracket of $\mathcal{M}^y$ is a sum of local times with weights that decay like $|x - y|^{2-2d}$; in the plane each dyadic annulus contributes the same order, which gives an extra factor $\log r_n$. For functions $f_1, f_2: \mathbb{Z}^d \rightarrow \mathbb{R}$ with bounded nearest-neighbor increments, let

$$(8.8) \quad \Gamma(f_1, f_2)(x) := \frac{1}{2d} \sum_{|e|=1} (f_1(x+e) - f_1(x))(f_2(x+e) - f_2(x)) - \Delta f_1(x) \Delta f_2(x),$$

which is the conditional covariance of the increments of $\mathcal{M}^{f_1}$ and $\mathcal{M}^{f_2}$ at a departure from x other than the first. As in Section 3.1, $g_y = g(\cdot - y)$ and G is the Green function.

Theorem 8.3 (Local times at fixed sites). *Let Ψ be the Euclidean norm, and let $0 < \varepsilon < 1/d$. Then the following hold.*

(i) Central limit theorem: for every $y \in \mathbb{Z}^d$, as $n \rightarrow \infty$,

$$(8.9) \quad \begin{aligned} \frac{\ell_n(y) - 2d\varepsilon(r_n - |y|)}{\sqrt{r_n \log r_n}} &\xrightarrow{d} \mathcal{N}\left(0, \frac{16\varepsilon}{\pi}\right) && \text{if } d = 2, \\ \frac{\ell_n(y) - 2d\varepsilon(r_n - |y|)}{\sqrt{r_n}} &\xrightarrow{d} \mathcal{N}(0, 2d\varepsilon(2G(0) - 1)) && \text{if } d \geq 3. \end{aligned}$$

(ii) Law of the iterated logarithm: for every $y \in \mathbb{Z}^d$ and each choice of sign, almost surely,

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{\pm(\ell_n(y) - 2d\varepsilon(r_n - |y|))}{\sqrt{2r_n \log r_n \log \log n}} &= \left(\frac{16\varepsilon}{\pi}\right)^{1/2} && \text{if } d = 2, \\ \limsup_{n \rightarrow \infty} \frac{\pm(\ell_n(y) - 2d\varepsilon(r_n - |y|))}{\sqrt{2r_n \log \log n}} &= (2d\varepsilon(2G(0) - 1))^{1/2} && \text{if } d \geq 3. \end{aligned}$$

(iii) Joint limits: for all $y_1, \dots, y_k \in \mathbb{Z}^d$, the convergence in (8.9) holds jointly for $y = y_1, \dots, y_k$, and the limit is a centered Gaussian vector whose covariance matrix has (i, j) entry

$$(8.10) \quad \begin{cases} 16\varepsilon/\pi, & d = 2, \\ 2d\varepsilon(2G(y_i - y_j) - \mathbf{1}_{\{y_i = y_j\}}), & d \geq 3 \end{cases}.$$

In the plane the limits at different sites are equal; this statement concerns finitely many fixed sites, whereas the maximum in Theorem 1.3 (iii) is over the sites x with $|x| \leq r_n^{1/2}$, a set that grows with n .

Proof of Theorem 8.3. Step 1: The bracket. By (3.5), at the first departure from x the conditional covariance of the increments of $\mathcal{M}^y$ and $\mathcal{M}^z$ differs from $\Gamma(g_y, g_z)(x)$ by $O_{y,z}((1 + |x|)^{2-2d})$. By Theorem 1.1 (i), almost surely $A_n \subset B(0, Cr_n)$ for large n , and summing over A_n gives, almost surely for large n ,

$$(8.11) \quad \langle \mathcal{M}^y, \mathcal{M}^z \rangle_n = \sum_{x \in \mathbb{Z}^d} \ell_n(x) \Gamma(g_y, g_z)(x) + \begin{cases} O_{y,z}(\log r_n), & d = 2, \\ O_{y,z}(1), & d \geq 3 \end{cases}.$$

Step 2: The bracket in higher dimensions. For $d \geq 3$, the weights $\Gamma(g_y, g_z)$ are absolutely summable. Theorem 1.1 (ii), dominated convergence and summation by parts give, almost surely,

$$\frac{\langle \mathcal{M}^y, \mathcal{M}^z \rangle_n}{r_n} \longrightarrow 2d\varepsilon \sum_{x \in \mathbb{Z}^d} \Gamma(g_y, g_z)(x) = 2d\varepsilon (2G(y - z) - \mathbf{1}_{\{y=z\}}).$$

The boundary term in the summation by parts is $O(r^{2-d})$ by (3.4) and (3.5).

Step 3: The bracket in the plane. Let $d = 2$. By (3.4) and (3.5), the increments of g_y at x are $(2/\pi)e \cdot x/|x|^2 + O_y(|x|^{-2})$, and $\sum_{|e|=1} (e \cdot x)^2 = 2|x|^2$. So, for fixed y, z and as $|x| \rightarrow \infty$,

$$\Gamma(g_y, g_z)(x) = \frac{2}{\pi^2|x|^2} + O_{y,z}(|x|^{-3}), \quad \text{and hence} \quad \sum_{|x| \leq r} \Gamma(g_y, g_z)(x) = \frac{4}{\pi} \log r + O_{y,z}(1).$$

In (8.11), replacing $\ell_n(x)$ by $2d\varepsilon(r_n - |x|)_+$ changes the sum by $o(r_n \log r_n)$, since the uniform error in Theorem 1.1 (ii) is $o(r_n)$ and $\sum_{|x| \leq Cr_n} |\Gamma(g_y, g_z)(x)| = O(\log r_n)$. For $|x| < r_n$ write $2d\varepsilon(r_n - |x|)_+ = 2d\varepsilon r_n - 2d\varepsilon|x|$: the constant term $2d\varepsilon r_n = 4\varepsilon r_n$ contributes $(16\varepsilon/\pi)r_n \log r_n + O(r_n)$, and the term $-2d\varepsilon|x|$ contributes $O(r_n)$, since $\sum_{|x| \leq Cr_n} |x|(1 + |x|)^{-2} = O(r_n)$. Therefore

$$\frac{\langle \mathcal{M}^y, \mathcal{M}^z \rangle_n}{r_n \log r_n} \longrightarrow \frac{16\varepsilon}{\pi} \quad \text{almost surely.}$$

Step 4: Limit laws for the martingales. The bounded increments and the bracket limits verify the hypotheses of the martingale central limit theorem [Hall and Heyde, 1980, Corollary 3.1] and Stout's law [Stout, 1970]. The Cramér–Wold device gives the joint Gaussian limits for $(-\mathcal{M}_n^{y_i})_i$, including directions with zero limiting variance; Stout's law gives both signs of the almost-sure limits at each fixed site.

Step 5: The potential term. By (3.12),

$$\ell_n(y) - 2d\varepsilon(r_n - |y|) = (U_{D_n}(y) - 2d\varepsilon(r_n - |y|)) - \mathcal{M}_n^y + O(\log n).$$

Lemma 8.2 and Theorem 8.1 make the potential term and the error negligible on the normalizations for the central limit theorem and the law of the iterated logarithm. Step 4 therefore proves the theorem. $\square$

9. LOWER BOUNDS ON THE FLUCTUATIONS

We prove the lower bounds of Theorem 1.3, an upper bound on the deviations of the local times away from the boundary (Proposition 9.2), and logarithmic lower bounds on the radii (Proposition 9.4). The lower bounds in Theorem 1.3 (i), (iii) and (iv)(a) need martingale deviations of about $(\log n)^{1/2}$ standard deviations, which we obtain from exponential martingales (Lemma 9.1): for parts (i) and (iv)(a) by the Paley–Zygmund inequality, and for part (iii), where the deviation at a fixed site is typically smaller than the bound in part (iii) by a factor of order $(\log n)^{1/2}$ (Theorem 8.3), by a second-moment bound for the average of the exponential martingales at many well-separated sites.

Throughout this section $0 < \varepsilon < 1/d$, and Ω_n is the event of Section 5.1 with $p = 10$, so $\mathbb{P}(\Omega_n^c) = O(n^{-10})$ and the estimates of Sections 3–7 used below hold on Ω_n for large n . In particular, by Section 7.4, on Ω_n for large n ,

$$(9.1) \quad \max_{x \in \mathbb{Z}^d} |\ell_n(x) - 2d\varepsilon(r_n - |x|)_+| \leq Cr_n\lambda_n, \quad \text{where} \quad \lambda_n := r_n^{-1/2}(\log n)^{3/2}.$$

Lemma 9.1. *Let $0 < c_0 \leq C_0$. There exist $c(c_0, C_0) > 0$ and $C(c_0, C_0) < \infty$ with the following property. Let m and n be positive integers, let $b > 0$, $\delta \geq 0$ and $0 \leq \alpha \leq 1$, and let $S^1, \dots, S^m$ be real martingales with respect to a common filtration, with $S_0^i = 0$ and $|S_{t+1}^i - S_t^i| \leq b$ for $0 \leq t < n$. Let Ω be an event with $\mathbb{P}(\Omega) \geq 1 - \alpha$ on which $c_0 \leq \langle S^i \rangle_n \leq C_0$ for $1 \leq i \leq m$. Then, for every real $\beta \geq C$ with $\beta b \leq c$, the following hold.*

(i) *For $1 \leq i \leq m$,*

$$\mathbb{P}(S_n^i \geq c\beta) \geq \frac{1}{4}e^{-C\beta^2} - \alpha \quad \text{and} \quad \mathbb{P}(S_n^i \leq -c\beta) \geq \frac{1}{4}e^{-C\beta^2} - \alpha.$$

(ii) *If $|\langle S^i, S^j \rangle_n| \leq \delta$ on Ω for all $i \neq j$, and $\beta^2\delta + \beta^3b \leq 1$, then*

$$\mathbb{P}\left(\max_{1 \leq i \leq m} S_n^i < c\beta\right) \leq C(m^{-1}e^{C\beta^2} + \beta^2\delta + \beta^3b + e^{C\beta^2}\sqrt{\alpha}).$$

Proof. Let $(\mathcal{G}_t)_{0 \leq t \leq n}$ be the filtration, let τ be the least integer $0 \leq t < n$ such that $\langle S^i \rangle_{t+1} > 2C_0$ for some i , and let $\tau := n$ if there is none. Since brackets are predictable, τ is a stopping time. The martingales S^i stopped at τ have increments at most b and brackets at most $2C_0$ at time n , and on Ω they agree with S^i up to time n , because $\tau = n$ there. From now on S^i denotes the stopped martingale; this changes neither the hypotheses nor S_n^i on Ω . Let $V_{ij} := \langle S^i, S^j \rangle_n$, so that $|V_{ij}| \leq 2C_0$ by the Cauchy–Schwarz inequality. We take $c \leq \min\{1/4, c_0/8\}$ small and C large, both depending only on c_0 and C_0 .

Let Y be a random variable and $\mathcal{H}$ a σ -field with $\mathbb{E}(Y | \mathcal{H}) = 0$ and $|Y| \leq b_1$, and let $s \in \mathbb{R}$ satisfy $|s|b_1 \leq 1$. Taylor’s formula for the exponential and the logarithm gives, with an absolute implicit constant,

$$(9.2) \quad \log \mathbb{E}(e^{sY} | \mathcal{H}) = \frac{s^2}{2}\mathbb{E}(Y^2 | \mathcal{H}) + O(|s|^3b_1\mathbb{E}(Y^2 | \mathcal{H})).$$

Now let S be a martingale with respect to $(\mathcal{G}_t)$, with $S_0 = 0$, increments at most b_1 and $\langle S \rangle_n \leq 8C_0$, let $|s|b_1 \leq 1/2$, and let

$$K(s) := \sum_{t < n} \log \mathbb{E}(e^{s(S_{t+1} - S_t)} | \mathcal{G}_t) \quad \text{and} \quad Z(s) := \exp\{sS_n - K(s)\}.$$

By successive conditioning $\mathbb{E}Z(s) = 1$, and (9.2) gives $K(s) = s^2\langle S \rangle_n/2 + O(|s|^3b_1)$. Hence $K(2s) - 2K(s) \leq Cs^2$, and since $Z(s)^2 = Z(2s)\exp\{K(2s) - 2K(s)\}$, we get $\mathbb{E}Z(s)^2 \leq e^{Cs^2}$. We apply the definitions and bounds of this paragraph with $s = \beta$ to S^i , with $b_1 = b$, and to $S^i + S^j$ for $i \neq j$, whose bracket at time n is $V_{ii} + V_{jj} + 2V_{ij} \leq 8C_0$, with $b_1 = 2b$; this is possible because $\beta b \leq 1/4$. Let K_i, Z_i and K_{ij}, Z_{ij} be the corresponding values of $K(\beta)$ and $Z(\beta)$. Then

$$(9.3) \quad K_i = \frac{\beta^2}{2}V_{ii} + O(\beta^3b) \quad \text{and} \quad K_{ij} - K_i - K_j = \beta^2V_{ij} + O(\beta^3b),$$

and $\mathbb{E}Z_i = \mathbb{E}Z_{ij} = 1$, $\mathbb{E}Z_i^2 \leq e^{C\beta^2}$ and $\mathbb{E}Z_{ij}^2 \leq e^{C\beta^2}$.

We prove (i). If c is small enough, then on Ω the bound $V_{ii} \geq c_0$ and (9.3) give $K_i \geq c_0\beta^2/4$. If also C is large enough, then on Ω the inequality $Z_i \geq 1/2$ implies $\beta S_n^i \geq c_0\beta^2/4 - \log 2$, hence $S_n^i \geq c_0\beta/8 \geq c\beta$. By the Paley–Zygmund inequality, $\mathbb{P}(Z_i \geq 1/2) \geq (\mathbb{E}Z_i)^2/(4\mathbb{E}Z_i^2) \geq e^{-C\beta^2}/4$, so $\mathbb{P}(S_n^i \geq c\beta) \geq \mathbb{P}(\Omega \cap \{Z_i \geq 1/2\}) \geq e^{-C\beta^2}/4 - \alpha$. The same argument applied to $-S^i$ gives the second bound in (i).

We prove (ii). Let $i \neq j$. By (9.3), the exponent in $Z_i Z_j = Z_{ij} \exp\{K_{ij} - K_i - K_j\}$ is at most $\beta^2 \delta + C\beta^3 b$ on Ω , and at most $C\beta^2$ everywhere, because $|V_{ij}| \leq 2C_0$ and $\beta b \leq 1$. So the Cauchy–Schwarz inequality on Ω^c and $\beta^2 \delta + \beta^3 b \leq 1$ give

$$\mathbb{E}(Z_i Z_j) \leq e^{\beta^2 \delta + C\beta^3 b} + e^{C\beta^2} (\mathbb{E} Z_{ij}^2)^{1/2} \alpha^{1/2} \leq 1 + C(\beta^2 \delta + \beta^3 b) + e^{C\beta^2} \sqrt{\alpha}.$$

Let $Z := m^{-1} \sum_{i=1}^m Z_i$. Then $\mathbb{E} Z = 1$, and the bounds on $\mathbb{E} Z_i^2$ and on $\mathbb{E}(Z_i Z_j)$ give

$$\text{Var}(Z) = \frac{1}{m^2} \sum_{i,j=1}^m (\mathbb{E}(Z_i Z_j) - 1) \leq C(m^{-1} e^{C\beta^2} + \beta^2 \delta + \beta^3 b + e^{C\beta^2} \sqrt{\alpha}).$$

On Ω , if $Z \geq 1/2$, then $Z_i \geq 1/2$ for some i , and hence $\max_i S_n^i \geq c\beta$, as in the proof of (i). So Chebyshev's inequality gives $\mathbb{P}(\max_i S_n^i < c\beta) \leq 4 \text{Var}(Z) + \alpha$, which proves (ii), because $\alpha \leq \sqrt{\alpha}$. $\square$

9.1. Lower bounds on the radii and their difference in the plane. Up to additive constants, $R_{\text{in}}(n) \leq R_{\text{mom}}(n) \leq R_{\text{out}}(n)$, and by (8.1) the difference $R_{\text{mom}}(n) - r_n$ is, to first order, proportional to $\mathcal{Q}_n$. So Theorem 1.3 (i) needs deviations of $\mathcal{Q}_n$ of about $(\log n)^{1/2}$ standard deviations with probability at least n^{-p} . Lemma 9.1 (i) gives them, because on Ω_n the bracket of $\mathcal{Q}$ is close to the deterministic value $(4\pi\varepsilon/5)r_n^5$. For part (iv)(a) we apply Lemma 9.1 (i) to the first coordinate of the martingale (4.1), whose bracket lies between two constant multiples of n on the whole probability space, for part (iv)(b) the martingale law of the iterated logarithm to the same coordinate, and for part (iv)(c) the martingale central limit theorem to the martingale (4.1) itself.

Proof of Theorem 1.3 (i) and (ii). Step 1: Comparison of the radii. Comparison with unit cells gives, for every $r \geq 1$,

$$\sum_{x \in \mathbb{Z}^d, |x| < r} |x| = \frac{d\omega_d}{d+1} r^{d+1} + O(r^d).$$

Every lattice site x with $|x| < R_{\text{in}}(n)$ lies in D_n , hence in A_n , since the cells are disjoint and $x \in C_x$; also $A_n \subset B(0, R_{\text{out}}(n) + 1)$. So $\sum_{x \in A_n} |x|$ lies between the sums of $|x|$ over the lattice sites in these two balls, and $s^{d+1} - t^{d+1} \geq (s-t)s^d$ for $s \geq t \geq 0$ gives, for every path,

$$(9.4) \quad R_{\text{in}}(n) \leq R_{\text{mom}}(n) + C \quad \text{and} \quad R_{\text{out}}(n) \geq R_{\text{mom}}(n) - C.$$

Step 2: The laws of the iterated logarithm. By (9.4), $r_n - R_{\text{in}}(n) \geq r_n - R_{\text{mom}}(n) - C$ and $R_{\text{out}}(n) - r_n \geq R_{\text{mom}}(n) - r_n - C$. For $d = 2$ the right side of (8.3) is $(20\pi\varepsilon)^{-1/2}$, so (8.3) gives

$$\limsup_{n \rightarrow \infty} \frac{r_n - R_{\text{in}}(n)}{\sqrt{r_n \log \log n}} \geq \frac{\sqrt{2}}{\sqrt{20\pi\varepsilon}} = \frac{1}{\sqrt{10\pi\varepsilon}},$$

and the same bound for $R_{\text{out}}(n) - r_n$, which proves (1.4).

Step 3: The polynomial lower bounds. With K as in Section 5.1, $R_{\text{out}}(n) < Kr_n$ on Ω_n . Let $\widehat{\mathcal{Q}}$ be $\mathcal{Q}$ stopped when the walk first leaves $B(0, Kr_n)$. Its increments are at most $4Kr_n$, and on Ω_n it agrees with $\mathcal{Q}$ up to time n , so $\langle \widehat{\mathcal{Q}} \rangle_n = \langle \mathcal{Q} \rangle_n$ on Ω_n , and (8.4) gives $\langle \mathcal{Q} \rangle_n$. In the first sum of (8.4), replacing $\ell_n(x)$ by $2d\varepsilon(r_n - |x|)_+$ changes the sum by at most $Cr_n^5 \lambda_n$, by (9.1), because ℓ_n and $2d\varepsilon(r_n - |\cdot|)_+$ both vanish outside $B(0, Kr_n)$; replacing the resulting lattice sum by the integral in Step 1 of the proof of Theorem 8.1 changes it by $O(r_n^4)$; and the second sum in (8.4) is $O(r_n^4)$. Since the limit in (8.5) is $4\pi\varepsilon/5$ for $d = 2$, this gives, on Ω_n for large n ,

$$\left| \langle \widehat{\mathcal{Q}} \rangle_n - \frac{4\pi\varepsilon}{5} r_n^5 \right| \leq Cr_n^5 \lambda_n \leq \frac{2\pi\varepsilon}{5} r_n^5.$$

Let $c_1 > 0$ and C_1 be the constants of Lemma 9.1 for $c_0 = 2\pi\varepsilon/5$ and $C_0 = 6\pi\varepsilon/5$. Fix $p > 0$, choose $a_p > 0$ with $C_1 a_p^2 \leq \min\{p, 10\}/2$, and let $\beta := a_p (\log n)^{1/2}$. For large n , $\beta \geq C_1$ and $4K\beta r_n^{-3/2} \leq c_1$. So Lemma 9.1 (i), with $m = 1$, $S^1 = \widehat{\mathcal{Q}}/r_n^{5/2}$, $b = 4K r_n^{-3/2}$, $\Omega = \Omega_n$ and $\alpha = \mathbb{P}(\Omega_n^c) = O(n^{-10})$,

shows that each of the events $\{\mathcal{Q}_n \geq c_1 \beta r_n^{5/2}\}$ and $\{\mathcal{Q}_n \leq -c_1 \beta r_n^{5/2}\}$ has probability at least $n^{-\min\{p, 10\}/2}/4 - O(n^{-10})$ after intersection with Ω_n , since $\widehat{\mathcal{Q}}_n = \mathcal{Q}_n$ on Ω_n . On these intersections, by (8.1) and $|X_n|^2 \leq C r_n^2$,

$$\sum_{x \in A_n} |x| - \frac{n}{2\varepsilon} \geq c r_n^{5/2} (\log n)^{1/2} \quad \text{and} \quad \frac{n}{2\varepsilon} - \sum_{x \in A_n} |x| \geq c r_n^{5/2} (\log n)^{1/2},$$

respectively. Since $R_{\text{mom}}(n)^3 - r_n^3 = (3/(2\pi))(\sum_{x \in A_n} |x| - n/(2\varepsilon))$ and $R_{\text{mom}}(n) \leq C r_n$ on Ω_n , factoring the difference of cubes gives $R_{\text{mom}}(n) - r_n \geq c \sqrt{r_n \log n}$ on the first intersection and $r_n - R_{\text{mom}}(n) \geq c \sqrt{r_n \log n}$ on the second, so by (9.4) each event in (1.3) has probability at least $n^{-\min\{p, 10\}/2}/4 - O(n^{-10}) \geq n^{-p}$ for large n . $\square$

Proof of Theorem 1.3(iv). Let $d = 2$, let $V_n := \sum_{x \in A_n} u_x$, and let Z be the martingale (4.1), so that $Z_n = X_n + \varepsilon V_n$, $Z_0 = 0$ and the increments of Z are at most 2. As in Step 1 of the proof of Theorem 8.1, for every $q \in \mathbb{R}^2$ the increment of $q \cdot Z$ at time j has conditional variance $|q|^2/2 - \varepsilon^2 I_j(q \cdot u_{X_j})^2$, so

$$(9.5) \quad \langle q \cdot Z \rangle_n = \frac{|q|^2 n}{2} - \varepsilon^2 \sum_{x \in A_n} (q \cdot u_x)^2.$$

Let $Y_n := e_1 \cdot Z_n$, so that $Y_n = (X_n)_1 + \varepsilon (V_n)_1$ and $n/2 - \varepsilon^2 |A_n| \leq \langle Y \rangle_n \leq n/2$.

Step 1: The sum over the range. Every lattice site x with $|x| < R_{\text{in}}(n)$ lies in A_n , and the sum of u_x over these sites vanishes, by the symmetry $x \mapsto -x$. So V_n is a sum of unit vectors over the sites $x \in A_n$ with $R_{\text{in}}(n) \leq |x| \leq R_{\text{out}}(n)$, whose cells lie in the annulus $\{R_{\text{in}}(n) - 1 \leq |v| \leq R_{\text{out}}(n) + 1\}$. Hence, for every path,

$$|V_n| \leq C(R_{\text{out}}(n) + 1)(R_{\text{out}}(n) - R_{\text{in}}(n) + 2).$$

Suppose moreover that $R_{\text{in}}(n) \geq r_n/2$ and $R_{\text{out}}(n) \leq 2r_n$, and let e be a unit vector. Then $|u_x - u_v| \leq C/r_n$ for these sites x and $v \in C_x$, so comparison with the integral of $(e \cdot u_v)_+$ over the annulus, whose angular factor is $\int_0^{2\pi} (\cos \theta)_+ d\theta = 2$, gives $e \cdot V_n \leq R_{\text{out}}(n)^2 - R_{\text{in}}(n)^2 + C r_n$; taking for e the direction of V_n gives

$$(9.6) \quad |V_n| \leq R_{\text{out}}(n)^2 - R_{\text{in}}(n)^2 + C r_n.$$

Step 2: The polynomial lower bound. With K as in Section 5.1, $|(X_n)_1| \leq R_{\text{out}}(n) < K r_n$ on Ω_n . Since $\varepsilon < 1/2$ and $|A_n| \leq n$, the martingale $S := Y/\sqrt{n}$ has bracket at time n between $c_0 := 1/2 - \varepsilon^2 > 0$ and $C_0 := 1/2$ on the whole probability space, and increments at most $b := 2n^{-1/2}$. Let $c_1 > 0$ and C_1 be the constants of Lemma 9.1 for these c_0 and C_0 , fix $p > 0$, choose $a_p > 0$ with $C_1 a_p^2 \leq \min\{p, 10\}/2$, and let $\beta := a_p (\log n)^{1/2}$. For large n , $\beta \geq C_1$ and $\beta b \leq c_1$, so Lemma 9.1 (i), with $m = 1$, the whole probability space in place of Ω and $\alpha = 0$, gives $\mathbb{P}(Y_n \geq c_1 \beta \sqrt{n}) \geq n^{-\min\{p, 10\}/2}/4$. On the intersection of this event with Ω_n , for large n ,

$$\varepsilon |V_n| \geq Y_n - |(X_n)_1| \geq c_1 a_p \sqrt{n \log n} - K r_n \geq \frac{c_1 a_p}{2} \sqrt{n \log n},$$

and Step 1 gives $R_{\text{out}}(n) - R_{\text{in}}(n) \geq c \sqrt{n \log n}/r_n - 2 \geq c \sqrt{r_n \log n}$, because $n = (4\pi\varepsilon/3)r_n^3$ by (1.1). Since $\mathbb{P}(\Omega_n^c) = O(n^{-10})$, this intersection has probability at least $n^{-\min\{p, 10\}/2}/4 - O(n^{-10}) \geq n^{-p}$ for large n , which proves (1.6).

Step 3: The iterated logarithm. By Theorem 1.1, almost surely $|A_n| = O(r_n^2) = o(n)$ and $R_{\text{in}}(n) + R_{\text{out}}(n) \sim 2r_n$. Hence $\langle Y \rangle_n/n \rightarrow 1/2$, and Stout's law for the martingale with bounded increments Y gives $\limsup Y_n/\sqrt{n \log \log n} = 1$. Since $(X_n)_1 = o(\sqrt{n})$, the identity $Y_n = (X_n)_1 + \varepsilon (V_n)_1$ and (9.6) give

$$\limsup_{n \rightarrow \infty} \frac{R_{\text{out}}(n) - R_{\text{in}}(n)}{\sqrt{r_n \log \log n}} \geq \frac{1}{2\varepsilon} \sqrt{\frac{4\pi\varepsilon}{3}} = \sqrt{\frac{\pi}{3\varepsilon}}.$$

Step 4: Small differences of the radii. Theorem 1.1 and (9.5) give $\langle q \cdot Z \rangle_n / n \rightarrow |q|^2 / 2$ for every q . The bounded increments, the martingale central limit theorem and the Cramér–Wold device give $Z_n / \sqrt{n} \xrightarrow{d} N$, where N is a centered Gaussian vector with covariance matrix $1/2$ times the identity. Thus $\mathbb{P}(|N| \leq t) = 1 - e^{-t^2}$ for $t \geq 0$.

By (9.6),

$$|Z_n| \leq \varepsilon (R_{\text{out}}(n) - R_{\text{in}}(n)) (R_{\text{out}}(n) + R_{\text{in}}(n)) + Cr_n.$$

Since $R_{\text{out}}(n) + R_{\text{in}}(n) \sim 2r_n$ and $\sqrt{n} = (4\pi\varepsilon/3)^{1/2} r_n^{3/2}$, the event $\{R_{\text{out}}(n) - R_{\text{in}}(n) \leq a\sqrt{r_n}\}$ forces $|Z_n|/\sqrt{n} \leq (3\varepsilon/\pi)^{1/2} a + o(1)$ almost surely. The portmanteau theorem and continuity of the radial Gaussian distribution give

$$\limsup_{n \rightarrow \infty} \mathbb{P}(R_{\text{out}}(n) - R_{\text{in}}(n) \leq a\sqrt{r_n}) \leq \mathbb{P}(|N| \leq (3\varepsilon/\pi)^{1/2} a) = 1 - e^{-3\varepsilon a^2/\pi}.$$

□

9.2. Upper bounds on the local times away from the boundary. At sites y with $|y| \leq \theta r_n$, where $0 < \theta < 1$, the set $D_n \setminus B(0, R_{\text{in}}(n))$ has small potential, because it has small volume and lies at distance of order r_n from y ; only the error δ of (6.2) remains. In the plane the resulting bound improves Theorem 1.2 (iii).

Proposition 9.2. *Let Ψ be the Euclidean norm, let $0 < \varepsilon < 1/d$, and let $0 < \theta < 1$ and $p > 0$. There exists $C(d, \varepsilon, \theta, p) < \infty$ such that, for every integer $n \geq 2$, with probability at least $1 - Cn^{-p}$,*

$$(9.7) \quad \max_{\substack{y \in \mathbb{Z}^d \\ |y| \leq \theta r_n}} |\ell_n(y) - 2d\varepsilon(r_n - |y|)| \leq C \begin{cases} \sqrt{r_n} \log n, & d = 2, \\ \sqrt{r_n} \log n, & d \geq 3 \end{cases}.$$

The same bound holds for all sufficiently large n almost surely.

Proof. Intersect the events of Theorem 1.2 and of Section 5.1, with exponent $\max\{p, 2\}$ in both, and let $E = D_n \setminus B(0, R_{\text{in}}(n))$ as in Section 6. For large n , every y with $|y| \leq \theta r_n$ lies at distance at least $c_\theta r_n$ from E , and Theorem 1.2 (i) and (ii) give $|E| \leq Cr_n^d q_n$; hence

$$|U_E(y)| \leq C_\theta |E| r_n^{1-d} \leq C_\theta r_n q_n.$$

By Theorem 1.2 (i), $|R_{\text{in}}(n) - r_n| \leq Cr_n q_n$, so (1.10) gives, uniformly over $|y| \leq \theta r_n$,

$$(9.8) \quad |U_{D_n}(y) - 2d\varepsilon(r_n - |y|)| \leq C_\theta r_n q_n.$$

Since $r_n q_n \leq \delta$, with δ as in (6.2), the bounds (6.2) and (9.8) prove (9.7) for large n , and the Borel–Cantelli lemma gives the almost-sure statement. □

9.3. Lower bounds on the local times near the origin. By (3.12) and (9.8), on Ω_n the deviation $\ell_n(y) - 2d\varepsilon(r_n - |y|)$ at a site y with $|y| \leq r_n/2$ equals $-\mathcal{M}_n^y$ up to $O(r_n q_n + \log n)$. So it suffices to find, among about $r_n^{1/8}$ sites at mutual distances at least $r_n^{1/4}$, one at which $-\mathcal{M}_n^y$ exceeds a multiple of $(\log r_n)^{1/2}$ standard deviations. In the plane the martingales at nearby sites are strongly correlated (see Step 3 of the proof of Theorem 8.3), so we use differences of martingales at two sites at distance about $r_n^{1/8}$. The normalized martingales then have brackets bounded above and below and cross brackets of order at most $r_n^{-1/4}$ (Lemma 9.3), and Lemma 9.1 (ii) applies.

Recall the martingales $\mathcal{M}^f$ of Section 3.1 and the function $\Gamma(f_1, f_2)$ of (8.8). Let $\sigma_n := (r_n \log r_n)^{1/2}$ and $k := \lfloor r_n^{1/8} \rfloor$ when $d = 2$, and $\sigma_n := r_n^{1/2}$ when $d \geq 3$. Let $m := \lfloor r_n^{1/8} \rfloor$, and for $1 \leq i \leq m$ let

$$y_i := i \lceil r_n^{1/4} \rceil e_1, \quad f_i := \begin{cases} g_{y_i + k e_1} - g_{y_i}, & d = 2, \\ g_{y_i}, & d \geq 3 \end{cases} \quad \text{and} \quad S^i := -\frac{\mathcal{M}^{f_i}}{\sigma_n}.$$

Then $|y_i| \leq m \lceil r_n^{1/4} \rceil \leq r_n^{1/2} / 2$ for large n .

Lemma 9.3. *Let Ψ be the Euclidean norm, and let $0 < \varepsilon < 1/d$. There exist $0 < c_0(d, \varepsilon) \leq C_0(d, \varepsilon)$ and $C(d, \varepsilon) < \infty$ such that, for all sufficiently large n , on Ω_n , for $1 \leq i \leq m$ and $1 \leq j \leq m$ with $j \neq i$,*

$$c_0 \leq \langle S^i \rangle_n \leq C_0 \quad \text{and} \quad |\langle S^i, S^j \rangle_n| \leq Cr_n^{-1/4}.$$

Proof. Step 1: Sums of $\Gamma(f_i, f_j)$. For $d \geq 3$, the second bound in (3.5) gives

$$\sum_{x \in \mathbb{Z}^d} |\Gamma(f_i, f_j)(x)| \leq C \quad \text{and} \quad \sum_{x \in \mathbb{Z}^d} \min\{|x|, r_n\} |\Gamma(f_i, f_j)(x)| \leq C(r_n^{1/2} + \log r_n).$$

Bound the product of two increments by half the sum of their squares, and use $|x| \leq |x - y_i| + r_n^{1/2}$. By (3.5), $\max_{|e|=1} |f_i(x+e) - f_i(x)|^2 \leq C(1 + |x - y_i|)^{2-2d}$. The right side has bounded sum over all x ; its sum weighted by $|x - y_i|$ over $|x - y_i| \leq r_n$ is $O(\log r_n)$ when $d = 3$ and $O(1)$ when $d \geq 4$; and its sum over $|x - y_i| > r_n$ is $O(r_n^{2-d})$.

For $d = 2$, (3.4) and (3.5) give

$$\max_{|e|=1} |f_i(x+e) - f_i(x)| \leq \begin{cases} C((1 + |x - y_i|)^{-1} + (1 + |x - y_i - ke_1|)^{-1}), & |x - y_i| \leq 3k, \\ Ck(1 + |x - y_i|)^{-2}, & |x - y_i| > 3k \end{cases}.$$

The sum over x of the square of the right side is $O(\log r_n)$, and the same sum weighted by $|x - y_i|$ is at most $Ck \log r_n$, so

$$\sum_{x \in \mathbb{Z}^2} |\Gamma(f_i, f_j)(x)| \leq C \log r_n \quad \text{and} \quad \sum_{x \in \mathbb{Z}^2} \min\{|x|, r_n\} |\Gamma(f_i, f_j)(x)| \leq Cr_n^{1/2} \log r_n.$$

The finitely supported products $\Delta f_i \cdot \Delta f_j$ satisfy the same bounds.

Step 2: The brackets. Let

$$\Sigma_{ij} := \frac{2d\varepsilon r_n}{\sigma_n^2} \sum_{x \in \mathbb{Z}^d} \Gamma(f_i, f_j)(x).$$

We show that $|\langle S^i, S^j \rangle_n - \Sigma_{ij}| \leq C\lambda_n$ on Ω_n for $1 \leq i, j \leq m$; note that $\lambda_n \leq r_n^{-1/4}$ for large n . The bracket $\langle S^i, S^j \rangle_n$ equals $\langle \mathcal{M}^{f_i}, \mathcal{M}^{f_j} \rangle_n / \sigma_n^2$. At a first departure from x , the conditional covariance of the increments of $\mathcal{M}^{f_i}$ and $\mathcal{M}^{f_j}$ differs from $\Gamma(f_i, f_j)(x)$ by at most a constant times the product of the largest increments of f_i and f_j at x ; on Ω_n these differences sum over $x \in A_n$ to $O(1)$ when $d \geq 3$ and $O(\log r_n)$ when $d = 2$. In $\sum_x \ell_n(x) \Gamma(f_i, f_j)(x)$, replace $\ell_n(x)$ first by $2d\varepsilon(r_n - |x|)_+$, which by (9.1) changes the sum by at most $C r_n \lambda_n \sum_x |\Gamma(f_i, f_j)(x)|$, and then by the constant $2d\varepsilon r_n$, which changes it by at most $2d\varepsilon \sum_x \min\{|x|, r_n\} |\Gamma(f_i, f_j)(x)|$. With Step 1, this proves the bound $|\langle S^i, S^j \rangle_n - \Sigma_{ij}| \leq C\lambda_n$.

Step 3: The covariances. For $d \geq 3$, Step 2 of the proof of Theorem 8.3 gives $\Sigma_{ij} = 2d\varepsilon(2G(y_i - y_j) - \mathbf{1}_{\{i=j\}})$. So $\Sigma_{ii} = 2d\varepsilon(2G(0) - 1) > 0$, and for $i \neq j$ the absolute value of Σ_{ij} is $4d\varepsilon G(y_i - y_j) = O(r_n^{(2-d)/4})$, by (3.4) and $|y_i - y_j| \geq r_n^{1/4}$. For $d = 2$, summation by parts on a ball of radius R , followed by $R \rightarrow \infty$, gives

$$\sum_{x \in \mathbb{Z}^2} \Gamma(f_i, f_j)(x) = 2(g(y_i - y_j + ke_1) + g(y_i - y_j - ke_1) - 2g(y_i - y_j)) - \sum_{x \in \mathbb{Z}^2} \Delta f_i(x) \Delta f_j(x),$$

where the last sum is 2 if $i = j$ and 0 otherwise, because the supports of Δf_i and Δf_j are disjoint for $i \neq j$. So $\Sigma_{ii} = 4\varepsilon(4g(ke_1) - 2) / \log r_n \rightarrow 4\varepsilon/\pi$ by (3.4), and for $i \neq j$ the right side is $O(k^2/|y_i - y_j|^2)$ by (3.4). Thus, in every dimension, the numbers Σ_{ii} lie between two positive constants for large n , and for all $i \neq j$,

$$(9.9) \quad |\Sigma_{ij}| \leq Cr_n^{-1/4}.$$

With Step 2, this proves the lemma. $\square$

Proof of Theorem 1.3 (iii). Step 1: The largest of the martingales. By (3.5), the increments of $\mathcal{M}^{f_i}$ are bounded by a constant, so those of S^i are at most $b := Cr_n^{-1/2}$. Let c_0, C_0 and $\delta_n := Cr_n^{-1/4}$ be the bounds of Lemma 9.3, and let $c_1 > 0$ and C_1 be the constants of Lemma 9.1 for these c_0 and C_0 . Fix $a > 0$ with $C_1 a^2 \leq 1/16$, and let $\beta := a(\log r_n)^{1/2}$. For large n , we have $\beta \geq C_1$, $\beta b \leq c_1$ and $\beta^2 \delta_n + \beta^3 b \leq 1$. So Lemma 9.1 (ii), with $\delta = \delta_n$, $\Omega = \Omega_n$ and $\alpha = \mathbb{P}(\Omega_n^c) = O(n^{-10})$, and the bounds $m^{-1} \leq 2r_n^{-1/8}$, $e^{C_1 \beta^2} \leq r_n^{1/16}$ and $r_n \leq n$ give

$$\mathbb{P}\left(\max_{1 \leq i \leq m} S_n^i < c_1 a (\log r_n)^{1/2}\right) \leq C(r_n^{-1/16} + r_n^{-1/4} \log r_n + r_n^{-1/2} (\log r_n)^{3/2} + r_n^{1/16-5}) \leq Cr_n^{-1/16}.$$

Step 2: From the martingales to the local times. On Ω_n the bound (9.8) with $\theta = 1/2$ holds for large n . If $S_n^i \geq c_1 a (\log r_n)^{1/2}$ for some i , then $-\mathcal{M}_n^{f_i} = \sigma_n S_n^i \geq c \sigma_n (\log r_n)^{1/2}$, and $r_n q_n + \log n$ is of smaller order than $\sigma_n (\log r_n)^{1/2}$. When $d \geq 3$, the representation (3.12) and (9.8) give $\ell_n(y_i) - 2d\varepsilon(r_n - |y_i|) \geq -\mathcal{M}_n^{f_i} - C(r_n q_n + \log n) \geq c\sqrt{r_n} \log n$. When $d = 2$, Dynkin's formula for f_i is the difference of (3.12) at $y_i + ke_1$ and at y_i , so (9.8) gives

$$|(\ell_n(y_i + ke_1) - 2d\varepsilon(r_n - |y_i + ke_1|)) - (\ell_n(y_i) - 2d\varepsilon(r_n - |y_i|))| \geq -\mathcal{M}_n^{f_i} - C(r_n q_n + \log n) \geq c\sqrt{r_n} \log n,$$

and hence $|\ell_n(y) - 2d\varepsilon(r_n - |y|)| \geq c/2\sqrt{r_n} \log n$ at $y = y_i$ or at $y = y_i + ke_1$. All these sites satisfy $|y| \leq r_n^{1/2}$. By Step 1, the probability that no index i has $S_n^i \geq c_1 a (\log r_n)^{1/2}$, plus $\mathbb{P}(\Omega_n^c)$, is at most $Cr_n^{-1/16} + Cn^{-10}$; since $r_n \asymp n^{1/(d+1)}$, this proves (1.5). $\square$

9.4. Logarithmic lower bounds for the radii. If D_n contains $B(0, r_n - h)$, or if $R_{\text{out}}(n) \leq r_n + h$ and $d \geq 3$, then the walk has visited many sites of an annulus of width of order h near the sphere of radius r_n ; in the second case this holds with high probability, by the martingale $\mathcal{Q}$ of (8.1). From each such site the walk leaves $B(0, r_n + h)$ along a straight path with conditional probability at least e^{-Ch} , so it follows none of these paths from cr_n^{d-1} first arrivals on disjoint time intervals with probability at most $\exp\{-cr_n^{d-1}e^{-Ch}\}$.

Proposition 9.4 (Lower bounds for the radii). *Let Ψ be the Euclidean norm, and let $0 < \varepsilon < 1/d$. There exist $c(d, \varepsilon) > 0$, $C(d, \varepsilon) < \infty$ and $h_0(d, \varepsilon) > 0$ such that the following hold.*

(i) Tail bounds: *for all sufficiently large n and every real h with $h_0 \leq h \leq r_n/8$,*

$$(9.10) \quad \mathbb{P}(R_{\text{in}}(n) \geq r_n - h \text{ and } R_{\text{out}}(n) \leq r_n + h) \leq \exp\{-cr_n^{d-1}e^{-Ch}\},$$

$$(9.11) \quad \mathbb{P}(R_{\text{out}}(n) \leq r_n + h) \leq \exp\{-cr_n^{d-1}e^{-Ch}\} + \exp\{-cr_n^{d-3}h^2\}.$$

(ii) Almost-sure bounds: *almost surely, for all sufficiently large n ,*

$$(9.12) \quad \max\{r_n - R_{\text{in}}(n), R_{\text{out}}(n) - r_n\} > c \log n,$$

and, if $d \geq 3$, also $R_{\text{out}}(n) - r_n > c \log n$.

In the plane the last term in (9.11) is not small when h has order $\log n$.

Proof. Step 1: Trials. By (2.1), every conditional nearest-neighbor transition probability is at least $p_0 := 1/(2d) - \varepsilon/2 > 0$. Let $T := \lceil 4h\sqrt{d} \rceil$. We show that, for every deterministic set $\Lambda \subset \{x \in \mathbb{Z}^d : |x| > r_n - 2h\}$ and every integer $k > T$,

$$(9.13) \quad \mathbb{P}(R_{\text{out}}(n) \leq r_n + h \text{ and } |A_n \cap \Lambda| \geq k) \leq (1 - p_0^T)^j, \quad \text{where } j := \left\lfloor \frac{k - T}{T + 1} \right\rfloor.$$

From each $x \in \Lambda$, the path of T steps in the direction $\text{sgn}(x_i)e_i$, where i is the smallest index with $|x_i| = \max_\ell |x_\ell|$, ends outside $B(0, r_n + h)$, because the derivative of $s \mapsto |x + s \text{sgn}(x_i)e_i|$ on $[0, \infty)$ is at least $|x_i|/|x| \geq d^{-1/2}$. A trial starts at each first arrival at a site of Λ that occurs by time $n - T$ and at or after the end of the preceding trial, and succeeds if the walk then follows this path from that site. If $|A_n \cap \Lambda| \geq k$, then at most T of the first arrivals at sites of $A_n \cap \Lambda$ occur after time $n - T$, and each trial interval contains at most $T + 1$ of them, so at least j trials start. Each trial starts at

a stopping time, succeeds with conditional probability at least p_0^T given the history at that time, and ends no later than the next trial starts; so, by induction, the probability that at least j trials start and the first j fail is at most $(1 - p_0^T)^j$. A successful trial ends outside $B(0, r_n + h)$ by time n , which proves (9.13).

Step 2: Two annuli. Let $\mathcal{S} := \{x \in \mathbb{Z}^d : r_n - 2h < |x| < r_n - h - 1\}$. After increasing h_0 , comparison with unit cells gives $|\mathcal{S}| \geq cr_n^{d-1}h$ for large n . On $\{R_{\text{in}}(n) \geq r_n - h\}$ every site of $\mathcal{S}$ lies in D_n , hence in A_n . So (9.13) with $\Lambda = \mathcal{S}$ and $k = |\mathcal{S}|$, for which $k > T$ and $j \geq cr_n^{d-1}$ for large n because $T \leq Ch$, bounds the left side of (9.10) by $(1 - p_0^T)^j \leq \exp\{-jp_0^T\} \leq \exp\{-cr_n^{d-1}e^{-Ch}\}$. In the same way, for $\Lambda := \{x \in \mathbb{Z}^d : r_n - 2h < |x| \leq r_n + h\}$ and every $c_1 > 0$,

$$(9.14) \quad \mathbb{P}(R_{\text{out}}(n) \leq r_n + h \text{ and } |A_n \cap \Lambda| \geq c_1 r_n^{d-1}h) \leq \exp\{-cr_n^{d-1}e^{-Ch}\},$$

where c also depends on c_1 .

Step 3: Many sites of A_n in Λ . Let $I(s) := \int_{B(0,s)} |v| dv$, so that $I(r_n) = n/(2\varepsilon)$. By Jensen's inequality, $|x| \leq \int_{C_x} |v| dv$ for every site x , so, after increasing h_0 so that $h_0 \geq \sqrt{d}$, the sites with $|x| \leq r_n - 2h$ contribute at most $I(r_n - 2h + \sqrt{d}) \leq I(r_n - h) \leq I(r_n) - c_2 r_n^d h$ to $\sum_{x \in A_n} |x|$, for a constant $c_2 > 0$. On $\{R_{\text{out}}(n) \leq r_n + h\}$ every site of A_n satisfies $|x| \leq r_n + h \leq 2r_n$, so $\sum_{x \in A_n} |x| \leq n/(2\varepsilon) - c_2 r_n^d h + 2r_n |A_n \cap \Lambda|$. Let $c_1 := c_2/4$. If moreover $|A_n \cap \Lambda| < c_1 r_n^{d-1}h$, then (8.1) gives $\mathcal{Q}_n \leq -\varepsilon c_2 r_n^d h + (r_n + h)^2 \leq -\varepsilon c_2 r_n^d h/2$ for large n , after increasing h_0 when $d = 2$. The martingale $\mathcal{Q}$ stopped at the first time τ with $|X_\tau| > r_n + h$ starts at 0, has increments at most $4(r_n + h) \leq 5r_n$, and equals $\mathcal{Q}_n$ at time n on $\{R_{\text{out}}(n) \leq r_n + h\}$, so Azuma's inequality and $n \leq Cr_n^{d+1}$ give

$$\mathbb{P}(R_{\text{out}}(n) \leq r_n + h \text{ and } |A_n \cap \Lambda| < c_1 r_n^{d-1}h) \leq \exp\left\{-\frac{(\varepsilon c_2 r_n^d h)^2}{200nr_n^2}\right\} \leq \exp\{-cr_n^{d-3}h^2\}.$$

With (9.14), this proves (9.11).

Step 4: The almost-sure bounds. Take $h = c \log n$ with $c(d, \varepsilon) > 0$ small enough. The right side of (9.10) is summable, as is that of (9.11) when $d \geq 3$. The Borel–Cantelli lemma gives (ii). $\square$

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